HPAL: A New Challenge for the Environment in Indonesia

A Focus on Nickel as a Raw Material of Electric Vehicle Batteries

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The Action for Ecology and People Emancipation (AEER)
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About:

The Action for Ecology and People Emancipation (AEER)

Action for Ecology and People Emancipation (AEER) is an Indonesian environmental organization founded in 2017, dedicated to addressing ecological and social justice issues, particularly those exacerbated by unsustainable industrial practices. AEER's mission is to empower communities impacted by environmental degradation and to advocate for policies that promote sustainability. The organization actively campaigns to reduce the nation's reliance on fossil fuels, particularly coal, while pushing for a transition to low-carbon energy alternatives. This commitment is reflected in its target to help reduce Indonesia's carbon emissions by 446 million tons of CO2 in the energy sector by 2030, aligning with the country's Enhanced Nationally Determined Contribution goals.

The organization collaborates with local and international partners, including community groups, government bodies, and other NGOs, to conduct research and advocacy campaigns. AEER's work is crucial in the context of Indonesia's growing nickel industry, which, while central to the global transition to electric vehicles, poses significant environmental and social challenges. By raising awareness, producing detailed research reports, and engaging in policy dialogue, AEER seeks to mitigate the adverse effects of mining and other industrial activities. The organization also emphasizes the importance of biodiversity conservation and the protection of critical ecosystems, working to ensure that economic development does not come at the expense of environmental integrity.

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Summary

Indonesia is a key country for the global nickel supply chain with 42.30 percent of global nickel reserves, representing 50 percent of total mined nickel and 42 percent of processed nickel in 2023. Even though Indonesia's processed nickel is mostly being used for stainless steel production, rapid growth of Chinese investment has turned Indonesia into the new center of nickel production. This is done through High-Pressure Acid Leaching (HPAL) technology that results in mixed hydroxide precipitate (MHP), a raw material of electric vehicle (EV) batteries.

Two mining companies, Sulawesi Cahaya Mineral (SCM) Ltd. and Hengjaya Mineralindo (HM) Ltd., and two smelter companies, Huayue Nickel Cobalt (HNC) Ltd. and QMB New Energy Materials (QMB) Ltd are part of the mixed hydroxide precipitate (MHP) supply chain production at the Indonesia Morowali Industrial Park (IMIP). Because these companies are part of large holdings in the integrated nickel business made up of Huayou Cobalt, GEM Co. Ltd, Merdeka Group, Nickel Industries, and Tsingshan Group, nickel production for electric vehicle batteries in Indonesia is characterized as monopolistic.

The role of Indonesia nickel production in the global green energy transition is ironic. Nickel mining operations are mandating deforestation and threatening biodiversity, causing water pollution, and triggering a lot of lawsuits against local communities. Nickel production generates low-paying jobs and use massive amounts of fossil fuels at captive coal power plants. HM, SCM, HNC, and QMB represent the filthy face of Indonesia's nickel supply chain.
To prevent more damages, the Indonesian government and nickel companies must implement high environmental and human rights standards as mandatory requirements in the nickel mining and smelting activities.

IMIP. Image Source: Google Earth

Key Takeaways

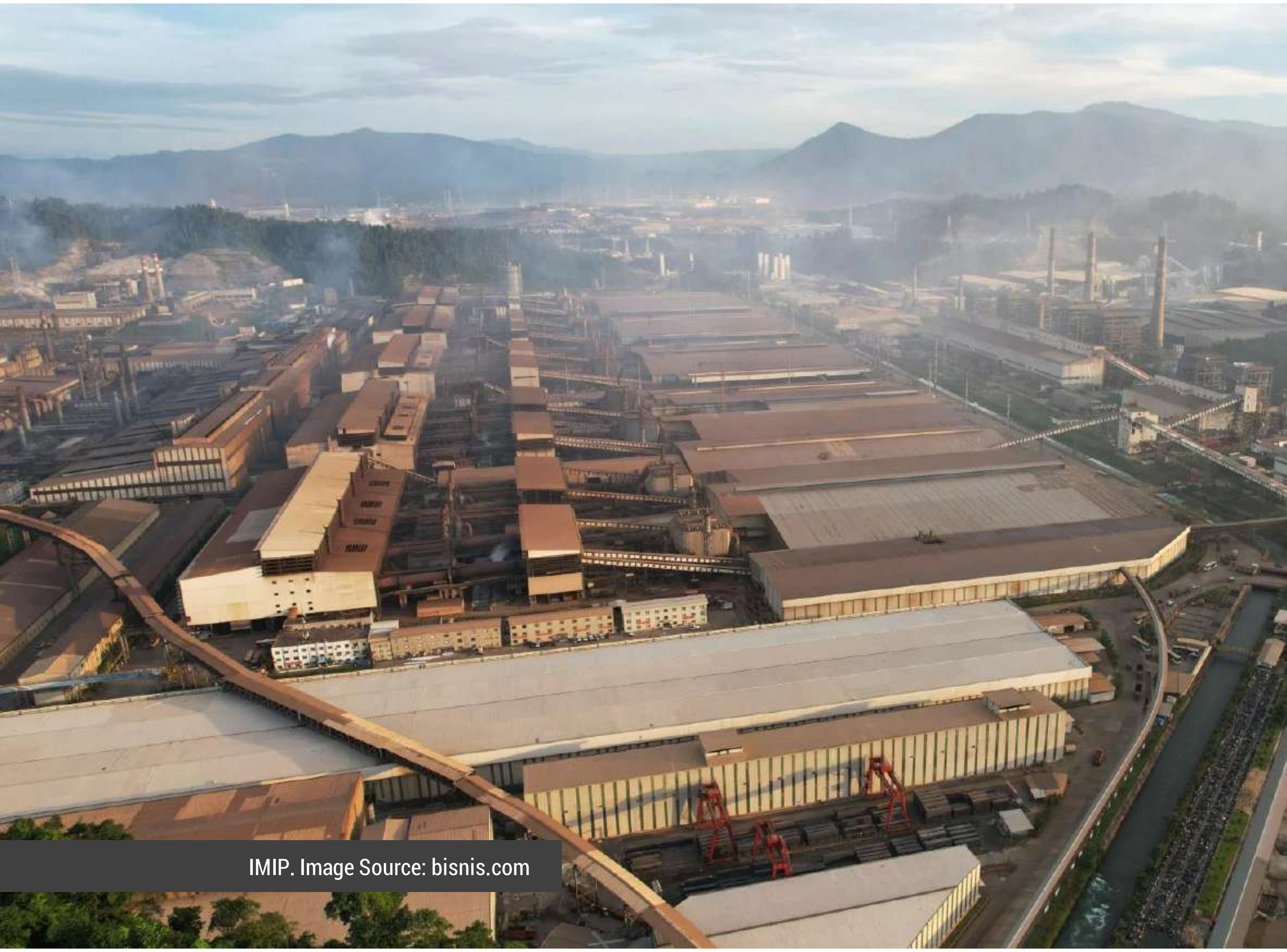
1. With 55 million tons or 42.30 percent of global nickel reserves, and the Indonesian government's downstream policies put Indonesia in the center of global nickel production, including nickel as the raw material for electric vehicle batteries. In 2023, 1.8 million tons of laterite ore was dredged from the earth's core in Central Sulawesi, East South Sulawesi, South Sulawesi, and North Maluku. In the same year, 3.36 million tons of global nickel production by Indonesia-China cooperation were estimated to provide 40.5 percent of processed nickel stocks globally. Given the global trend toward electric vehicles, Indonesia is the center of nickel production based on HPAL technology. This technology turns limonite ore into MHP as the raw material for EV batteries supply chain.

2. Companies in the nickel business are vertically and monopolistically involved in the supply production chain of MHP production, as demonstrated at IMIP. PT Hengjaya Mineralindo (HM), one of limonite ore suppliers for MHP production, is a part of integrated mining and processing projects in Indonesia. This company is jointly owned by Nickel Industries and Tsingshan Group. Sulawesi Cahaya Mineral (SCM) is a member of integrated mining and processing projects led by a cooperative partnership between Merdeka Group and Tsingshan Group. Huayue Nickel Cobalt (HNC), a MHP producer, is a division of Zhejiang Huayou Cobalt Co, Ltd., a giant Chinese metal company that has been expanding its business into nickel industry in Indonesia. QMB New energy Materials (QMB), a nickel refiner, is a subdivision of GEM Co., Ltd., a prominent Chinese green corporation that has been broadening its business into nickel-based new energy projects in Indonesia.

3. Unfortunately, in the name of clean energy, Indonesia is becoming the production site for filthy nickel that is used for the raw material for EV batteries globally. Deforestation that threatens biodiversity and causes water pollution is born from uncontrolled nickel mining operations. Wide ranging permits for nickel mining concessions and the establishment of new nickel-based industrial parks in the forest area have accelerated and expanded deforestation. Since 2013, the total amount of forest loss and potentially loss in the HM and SCM concessions areas are estimated at 6300 hectares. This is equal to 7680 international standard football fields. Mining operations have adversely affected local communities. MHP production using HPAL process is considered to be energy-intensive resulting in pollutant emission. It has happened as its activity leaned to off-grid coal power plants in IMIP. HNC and QMB are managing toxic tailing waste amounting to about

12.5 million tons per year. The use of filtered tailings technology is potentially harming underground water and surrounding water resources. It is also concerning, that these companies are operating filtered tailings facilities in a highly wet climate or high rainfall areas that are also categorized as disaster-prone areas (high frequency of earthquakes, landslides and floods).

Nickel production for clean energy is born of filthy business practices. HM and SCM undertake their mining activities by seizing local communities' lands. These companies expelled the local agricultural communities from their lands and subsequently criminalized local citizens. HNC and QMB employ many low-pay workers who work long hours, and lack health, safety and security standards.



IMIP. Image Source: bisnis.com

Recommendations

Given these results, the Indonesia government must implement the following recommendations:

1. Prevent further deforestation by decreasing the nickel Mining Business License Area for Performing Production (WIUP-OP) owned by SCM and HM, prohibit new forest-land clearing for mine support or infrastructure purposes, restrict the production quota for the nickel companies, which is driving forest-land clearing for mining activities, and evaluate the permitting of new industrial development in the forest area

2. Ensure all tailing facilities are managed by high security standards so as to protect society's safety and the environment. All regulations, including a dam ordinance, must be secured to ensure safe tailing facility operations

3. Revise Presidential Regulation Number 112/2022 to establish a new policy to force battery-grade nickel manufacturers to switch their coal-based electricity supply at processing operations to clean electricity. Indonesian government must prohibit new investment in battery-grade nickel industries that relies on a coal-based electricity supply.

4. Enforce regulations that require mining and processing companies to uphold high standards for health and safety for mine workers, including paying decent salaries and discontinuing long work hours. Punish nickel companies that disobey labor laws.

5. Ensure security forces are not involved in land disputes in mining zones and prevent all the methods of criminalization of local communities or farmers.

Furthermore, nickel mining companies must implement these recommendations:

1. Evaluate negative impacts on the environment and social aspects of its mining and smelting activities; prioritize transparent, detailed and actual reporting of public information.

2. Unlock opportunities for third party actors to independently monitor environmental and social impacts of nickel mining and smelting operations.

3. Discontinue use of coal-based power plants for HPAL operation and switch to clean energy electricity.

4. Follow the safety guidance for dams developed by International Commission on Large Dams (ICOLD) and responsibly manage tailing activities strictly following safety standards guidance according to national regulations (these include tailing and dam rules).
5. Ensure safety, health and security in the worksite, ensure decent salary for workers, decrease long working hours, implement fair action in labors disputes and allow laborers to independently build labor unions.

6. Ensure nickel companies' operations are bringing direct, positive impacts to nearby populations by increasing the local community's quality of life in social, economic and environmental aspects, not the contrary.

Abbreviations

ANI	:	Angel Nickel Industry
ANTAM	:	Aneka Tambang
AJB	:	Anugerah Jaya Buana
BKSDA	:	*Balai Konservasi Sumber Daya Alam* - Natural Resources Conservation Agency
BSI	:	Bukit Smelter Indonesia
CSI	:	Cahaya Smelter Indonesia
CATL	:	Contemporary Amperex Technology Co. Limited.
ENC	:	Excelsior Nickel Cobalt
ESDM	:	*Energi dan Sumber Daya Mineral* – Energy and Mineral Resources
ESG	:	ESG New Energy Material
EV	:	Electric Vehicle
GEM	:	Green Eco-Manufacture, GEM Co.Ltd.
HM	:	Hengjaya Mineralindo
HNC	:	Huayue Nickel Cobalt
HNI	:	Hengjaya Nickel Industry
HPAL	:	High-Pressure Acid Leaching
HPL	:	Halmahera Persada Lygend
HNMI	:	Huaneng Metal Industry
IDR	:	Indonesia Rupiah
IHIP	:	Indonesia Huabao Industrial Park
IKIP	:	Indonesia Konawe Industrial Park
IMIP	:	Indonesia Morowali Industrial Park
IPIP	:	Indonesia Pomalaa Industrial Park
IPPKH	:	*Izin Pinjam Pakai Kawasan Hutan* - Borrow-to-Use Forestry Permit
IUCN	:	International Union for Conservation of Nature
IUP-OP	:	*Izin Usaha Pertambangan Operasi Produksi* – Mining Business License for Performing Production
IUPK	:	*Izin Usaha Pertambangan Khusus* – Special Mining Business License
IWIP	:	Indonesia Weda Bay Industrial Park
KNI	:	Kolaka Nikel Indonesia
KK	:	*Kontrak Karya* – Contract of Work
LHK	:	*Lingkungan Hidup dan Kehutanan* - Envionment and Forestry
MBMA	:	Merdeka Battery Materials
MED	:	Merdeka Energi Industri
MHP	:	Mixed Hydroxide Precipitate
MIA	:	Merdeka Industri Anantha
MIN	:	Merdeka Industri Mineral

MMI	:	Merdeka Mega Industri
MTI	:	Merdeka Tsingshan Indonesia
MW	:	Megawatt
OESBF	:	Oxygen-Enriched Side-Blown Furnace
ONC	:	Obi Nickel Cobalt
ONI	:	Oracle Nickel Industry
QMB	:	Qing Mei Bang New Energy Materials
RKEF	:	Rotary-Kiln Electric Furnace
RMB	:	Renminbi
RNI	:	Ranger Nickel Industry
SCM	:	Sulawesi Cahaya Mineral
SPIM	:	*Serikat Pekerja Industrial Morowali* – Morowali Industrial Labor Union
SPN	:	*Serikat Pekerja Nasional* - National Labor Union
tCO2/tNi	:	ton carbon dioxide per ton nickel
t/y	:	ton/year
USD	:	United States Dollar
VDNIP	:	Virtue Dragon Nickel Industry Park
WIUP-OP	:	*Wilayah Izin Usaha Pertambangan Operasi Produksi* - Mining Business License Area for Performing Production
WKM	:	Wana Kencana Mineral
ZHN	:	Zhao Hui Nickel

FIRST SECTION
THE DEVELOPMENT OF THE INDONESIAN NICKEL INDUSTRY

1. Indonesian Nickel Production

Global nickel resources are estimated at around 350 million tons. This number includes 55 percent laterite ore and 35 percent sulfide ore. Indonesia's laterite reserve is estimated at 55 million tons or 42.30 percent of global nickel reserves.[1] Of the laterite reserve, 70 percent of the reserves are limonite ore (low-grade) and 30 percent reserves are saprolite (high-grade). These reserves are spreading around Central Sulawesi, Southeast Sulawesi, South Sulawesi, North Maluku, and West Papua.

Indonesia has become a key nickel player in the global mining industry. In 2023, 1.8 million tons of Indonesia nickel mining production was contributing 50 percent of global nickel mining output.[2] It was estimated that 7.3 million tons of nickel metal had been mined from Indonesia's soil between 2015-2023 period (see Figure 1). The largest mine sites for nickel are located in North Maluku, Southeast Sulawesi, Central Sulawesi, and South Sulawesi. There are 329 companies holding mining permits – Mining Business Licenses (IUP) and Contract of Work (KK) – that cover 836,000 hectares of land across the country.[3]

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1 U.S. Geological Survey, 2024, Mineral commodity summaries 2024: U.S. Geological Survey. Available at https://doi.org/10.3133/mcs2024. Accessed July 10, 2024

2 *Ibid*

3 Direktorat Jenderal Mineral dan Batubara Kementerian Energi dan Sumber Daya Mineral.2021. *Grand Strategi Mineral dan Batubara*. Jakarta: Direktorat Jenderal Mineral dan Batubara Kementerian Energi dan Sumber Daya Mineral.

Indonesia has become the center of global smelted nickel and refined nickel production since the smelting and refining production lines were established a decade ago. These production facilities were built in Central Sulawesi, Southeast Sulawesi, and North Maluku. In 2023, the world's total processed nickel production reached 3.36 million tons,[4] Indonesia was estimated to contribute 40.5 percent or around 1.36 million tons. Indonesia's contribution has the potential to rise into 43.9 percent by 2027, soaring sharply from 16.1 percent in 2019.[5] The rapid growth is factored by the presence of smelters that produce NPI (nickel pig iron), NPI-to-nickel matte conversion, and mixed hydroxide precipitate (MHP). An optimistic estimation predicts Indonesia manufactured 1.39 million tons of NPI in 2023, 1.57 million tons in 2024, and would be 1.7 million tons in 2025. The country produces nickel matte, including converted from NPI, with an estimated production capacity of 302,000 tons by 2024.[6] Indonesia already has High-Pressure Acid Leaching (HPAL) facilities that produce MHP with a production capacity of 355,000 tons per year.

Processed nickel and its derivative products dominate exports in Central Sulawesi, Southeast Sulawesi, and North Maluku. In these provinces, the total export value of the "iron and steel" coded harmonized system (HS-72) reached USD 24.9 billion in 2023, soaring from USD 185 million in 2015. HS-72 includes NPI/ferronickel, hot roll coil (HRC), cold roll coil (CRC), stainless steel slab, and carbon steel. Meanwhile, the total value of nickel exports (HS-75) from Central Sulawesi and North Maluku ascended from USD 311 million in 2021 to USD 5.5 billion in 2023. HS-75 includes MHP and nickel matte (see **Figure 2**).

These achievements result from mining downstream policies. Starting with Law Number 4 of 2009 concerning Mineral and Coal Mining which prohibits the export of nickel ore abroad. The prohibition is then contained in a series of Government Regulations and is technically implemented through a series of regulations of the Minister of Energy and Mineral Resources (ESDM) Indonesia issued since 2014.[7] The government also provides various investment facilities such as the construction of captive power plants in industrial estates,[8] tax holidays, designation as national strategic projects and national vital objects for nickel processing projects and nickel-based industrial parks.[9]

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4 INSG. 2024. "Nickel Market Observations for 2024 and 2025". *Press Release* 24 September.

5 Eri Silva. 2024. "Indonesian nickel production dominates commodity market". Available at https://www.spglobal.com/marketintelligence/en/news-insights/latest-news-headlines/indonesian-nickel-production-dominates-commodity-market-80242322. Accessed September 6, 2024.

6 Nornickel. 2024. "Quintessentially Nickel." Available at https://nornickel.com/upload/iblock/d18/iz37fmx2a1srcecotbrgm1mixwtzvwfm/2024_05_30-Quintessentially-Ni.pdf. Accessed June 30, 2024.

7 Minister of Energy and Mineral Resources Regulation (*Peraturan Menteri ESDM*) No. 01 of 2014 on the Enhancement of Mineral Added Value Through Mineral Processing and Refining Activities within the Country; Minister of Energy and Mineral Resources Regulation (*Peraturan Menteri ESDM*) No. 5 of 2017 on the Enhancement of Mineral Added Value Through Mineral Processing and Refining Activities within the Country. State Gazette of the Republic of Indonesia (*Berita Negara Republik Indonesia*) No. 98, 2017; Minister of Energy and Mineral Resources Regulation (*Peraturan Menteri ESDM*) No. 11 of 2019 on the Second Amendment to the Minister of Energy and Mineral Resources Regulation No. 2018 on Mining, Mineral, and Coal Business Operations.

8 Government Regulation (*Peraturan Pemerintah*) No. 142 of 2015 on Industrial Park.

9 Presidential Regulation (*Perpres*) No. 109 of 2020 on the Third Amendment to Presidential Regulation (Perpres) No. 3 of 2016 on the Acceleration of the Implementation of National Strategic Projects.

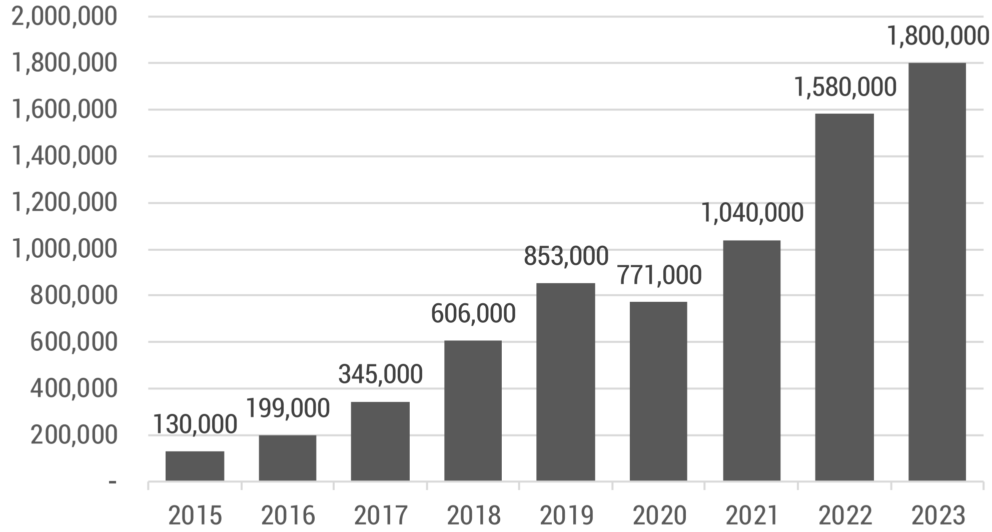

Figure 1. Indonesia: Nickel Mining Production 2015-2023 (in Thousand tons)

Source: Analyzed from USGS

Chinese capital takes advantage of downstream regulations by building nickel processing plants and supporting infrastructure such as captive power plants in Central Sulawesi, Southeast Sulawesi, and North Maluku. Tsingshan Group pioneered the first Rotary-Kiln Electric Furnace (RKEF) technology to produce low-cost NPI in Morowali in 2015. Since NPI is a feedstock for stainless steel, Chinese companies have built RKEF-Argon Oxygen Decarburization (AOD) integrated production line since 2016 to produce stainless steel so as to make an efficient industrial supply chain at the Indonesia Morowali Industrial Park (IMIP) in Morowali, Central Sulawesi and at the Virtue Dragon Nickel Industrial Park (VDNIP) in Konawe, Southeast Sulawesi. IMIP and VDNIP become the new hubs of the world's stainless-steel production.

Since the end of last decade, in the midst of the upward wave of the energy transition, Chinese giants such as Zhejiang Huayou Cobalt Co., Ltd., GEM Co., Ltd., Contemporary Amperex Technology Co., Ltd (CATL), Lygend Resources & Technology Co., Ltd., Nickel Industries Limited (NIC) and CNGR Advanced Material Co., Ltd., have been competing to invest in nickel projects in Indonesia. They are progressing nickel processing projects for raw materials for electric vehicle (EV) batteries by exploiting Indonesia's nickel abundance and forming partnership with leading domestic business groups like PT Trimegah Bangun Persada Tbk (NCKL) or Harita Group and PT Merdeka Copper Gold Tbk (MDKA). The impact of Chinese investment companies in the production of nickel for EV batteries has been the cementing of Indonesia as one of the key upstream actors in the global EV supply chain.

Indonesia's role will be essential considering the estimated nickel demand for EV batteries will reach 1.09 million tons by 2030, almost doubling from 2023. Currently, EV batteries with nickel-rich cathodes like NMC (nickel-manganese-cobalt) and NCA (nickel-cobalt-aluminum) continue dominating the market. Last year, the two common cathode types accounted for more than half of the global electric vehicle battery market.[10] Nickel is a key element in the production of precursor cathode active materials for EV batteries. The nickel-rich cathode in lithium-ion batteries has reversible capabilities and high energy density. The more nickel composition in the cathode, the higher the battery's ability to store energy, making the vehicle travel longer.

To ensure the security of nickel supply, the world's largest EV battery manufacturer, CATL not only purchases intermediate nickel products produced in Indonesia, but also directly invests in the mining and nickel processing industry of Indonesia. Through its subsidiary HongKong CBL Limited (HKCBL), CATL has controlled 49 percent of PT Sumberdaya Arindo, a nickel mining company that owns a Mining Business License for Performing Production covering an area of 14,421 hectares in Halmahera. In order to build an industrial estate and nickel-based battery processing industry in Indonesia, CATL has also controlled 60 percent of PT Feni Haltim's shares in Halmahera. Previously, both Arindo Resources and Feni Haltim were companies whose ownership was controlled by PT Aneka Tambang (Antam), a leading state-owned mining company.[11]

Ni-MH battery from Toyota NHW20 Prius. Image Source: electricvehiclesnews.com.

[10] IRENA. 2024. Critical Materials: Batteries for electric vehicles. Abu Dhabi: International Renewable Energy Agency; Transport & Environment. 2023. "Paving the way to cleaner nickel". Available at https://www.transportenvironment.org/uploads/files/2023_10_Briefing_Paving_way_cleaner_nickel-1.pdf. Accessed July 26, 2024; International Energy Agency (IEA). 2024. Global EV Outlook 2024: Moving towards increased affordability. Available at https://iea.blob.core.windows.net/assets/a9e3544b-0b12-4e15-b407-65f5c8ce1b5f/GlobalEVOutlook2024.pdf. Accessed July 26, 2024.

[11] PT Antam Tbk. 2024. 2023 Laporan Tahunan. Jakarta: PT Antam.

Instead of purchasing nickel-based batteries from EV battery makers, the world's leading EV manufacturers are involved in direct purchase agreements with processed nickel producers in Indonesia for the fabrication of their own EV batteries. Tesla said 13 percent of its nickel consumption comes from Indonesia and has made a nickel purchase agreement worth USD 5 billion with two processed nickel producers in Morowali.[12]

Volkswagen names Indonesia as one of the countries that supplies nickel to the company, which is the largest automaker in Europe. The supply seems to be delivered from Huayou Cobalt which has production facilities in IMIP and at Indonesia Weda Bay Industrial Park (IWIP) in North Maluku.[13] Ford Motor directly participates in a nickel processing project through cooperation with Huayou Cobalt and PT Vale Indonesia in Pomalaa, Southeast Sulawesi.[14]

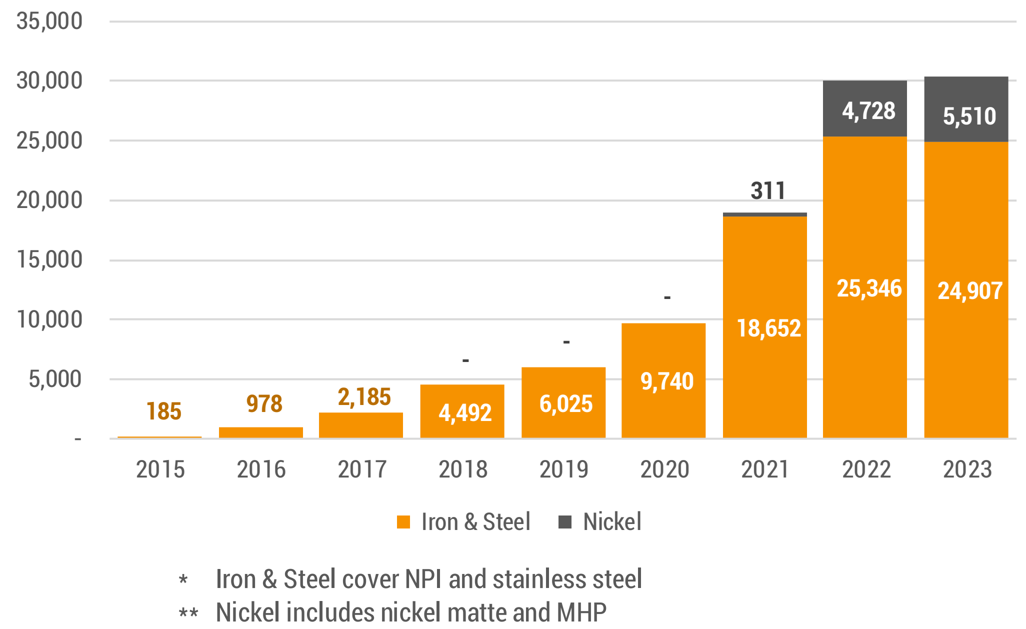

Figure 2. The Total Export of Iron, Steel and Nickel from Central Sulawesi, South East Sulawesi, and North Maluku in the 2015-2023 (in Million USD)

Source: Analyzed from BPS

[12] Fransiska Nangoy. 2022. "Indonesia says Tesla strikes $5 billion deal to buy nickel products". August 8. Available at https://www.reuters.com/business/autos-transportation/indonesia-says-tesla-strikes-5-bln-deal-buy-nickel-products-media-2022-08-08/ Accessed July 26, 2024.

[13] Volkswagen Group.2024. *Responsible Raw Materials Report 2023*. Available at https://uploads.vwmms.de/system/production/documents/cws/002/716/file_en/d4d4bc8b2aea8ace68435605a99ef6e9a9bbf973/2023_Volkswagen_Group_Responsible_Raw_Materials_Report_1.pdf?1719555968. Accessed July 26, 2024.

[14] Ford Newsroom. 2023. "PT Vale Indonesia and Huayou Sign Nickel Agreement with Ford Motor Co. Supporting Growth of the Global Sustainable EV Industry". March 30. Available at https://media.ford.com/content/fordmedia/fna/us/en/news/2023/03/30/pt-vale-indonesia-and-huayou-sign-nickel-agreement-with-ford-mot.html. Accessed July 26, 2024.

2. Technology Routes and the Nickel Production Value Chain in Indonesia

Nickel processing projects in Indonesia employ three technological routes to produce intermediate nickel products, including for raw materials for EV batteries (see **Figure 3**). The first is RKEF technology, which processes saprolite ore. This technology has been used in Indonesia for a long time. PT Inco Indonesia (now PT Vale Indonesia) has been working to produce nickel matte in Sorowako since 1978. PT Antam has also employed it for ferronickel production in Pomalaa since 1976. Since the arrival of Chinese investment, the usage of this pyrometallurgical technology has dramatically expanded. This began with the operation of the PT Sulawesi Mining Investment's smelter in Morowali in the mid-2015, which initially planned to produce NPI as a raw material for making stainless steel. At this time, the increasing demand for EV batteries triggered the use of this technology for the conversion of NPI-to-nickel matte, which can be refined into nickel sulfate.

The second technology is High-Pressure Acid Leaching (HPAL). With a smaller value amount of investment than previous HPAL projects in other parts of the world, Chinese investors have successfully built HPAL production lines with similar technology that produce MHP. This hydrometallurgical technology extracts nickel and cobalt from low-grade nickel ore. The technique uses dissolution process. The ore is filtered and mixed with water to form a slurry and then it is heated. Next it is put into an autoclave, or high-pressure vessel. Sulfuric acid is added and the mixture is heated to a high temperature and pressure so that it dissolves the nickel and cobalt. The nickel and cobalt-rich solution is then cooled and neutralized with an alkaline solution that produces MHP. Like nickel matte, this intermediate nickel is converted into nickel sulfate, as has been produced by PT Halmahera Persada Lygend on Obi Island, North Maluku. Nickel sulfate is used to manufacture EV battery precursors. Precursors are chemical compounds consisting of nickel, cobalt, manganese, and aluminum, key components for cathodes, one of the important materials in lithium batteries.

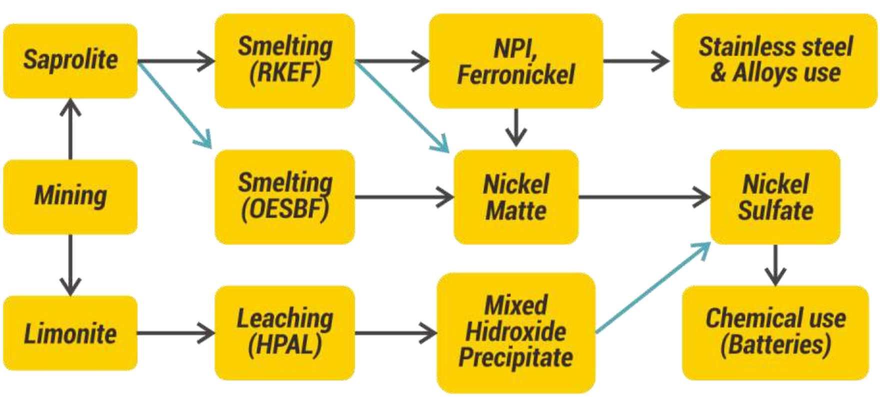

Figure 3. Technology Routes and Nickel Production Value Chain in Indonesia

Finally, the Chinese metal companies also successfully applied the Oxygen-Enriched Side-Blown Furnace (OESBF) technology to process nickel matte from saprolite ore. This technology enriches the existing methods of nickel processing to produce nickel-based ingredients for EV batteries materials. This technology was first operated in Indonesia at IMIP in October 2022 which is run by PT Zhongtsing New Energy, a subsidiary of CNGR Advanced Material, a world-class producer of lithium-ion battery cathode precursor.[15] It has been claimed that the OESBF technology has advantages over other laterite nickel ore processing technologies in terms of energy input. The technology consumes energy more efficiently and thus reduces carbon emissions. Another advantage is that this technology can adjust to raw materials inputs, such as low grade, moisture content, and particle size of laterite ore. Slag, a by-product of nickel smelting with this technology, is also claimed to be used as a building materials without the risk of environmental pollution.[16]

Copper-Cathode-Refinery-EPC4. Image Source: topmie.com

[15] CNGR. 2023. "Year End Review: Ten Significant Events of CNGR in 2022." Available at http://www.cngrgf.com.cn/en-US/gsxw/1014.html. Accessed July 27, 2024.

[16] CNGR. 2024. "More Than 2,000 Tons of Nickel Matte!The First Batch of Nickel Matte from CNGR Morowali Industrial Base Arrived at Qinzhou Port", Available at http://www.cngrgf.com.cn/en-US/gsxw/1242.html. Accessed July 27, 2024.

3. Environment and Human Rights

Given the rapid and massive growth, the nickel industry in Indonesia has several issues that are inevitable. Indonesia's nickel Industry led by Chinese investment is considered to produce "dirty nickel" for several reasons.

Deforestation. Nickel mining has become one of the main drivers of deforestation in Indonesia. Mining concessions are generally located in the forest areas and open pit mining requires clearing the forest. According to Mighty Earth, deforestation from nickel mining activities in Indonesia reached 75,000 hectares.[17] Previously, *Tempo* reported that around 500,000 hectares of forests in Indonesia's two main nickel-producing provinces, Central Sulawesi and Southeast Sulawesi, were depleted or threatened to run out in the 2011-2020 period.[18]

Nickel Mining in Morowali. Image Source: Dimas Ardian / Bloomberg.com

[17] Mighty Earth. 2024. *From Forest to Electric Vehicles.* Available at https://mightyearth.org/wp-content/uploads/2024/05/FromForeststoEVs.pdf. Accessed September 22, 2024.

[18] Erwan Hermawan. 2022. "Tentakel Tambang Nikel". *Tempo*, January 31-February 6.

Deforestation mainly occurs legally because it is based on Borrow-to-Use Forest Permit (IPPKH) issued by the Minister of Environment and Forestry (LHK). In Morowali between 2010 - 2021, the government issued 24 Borrow-to-Use Forest Permits covering an area of 11,942.43 hectares to 17 nickel companies for mining operation and infrastructure development.[19] However, even without the Borrow-to-Use Forest Permits, miners also rampantly clear forest lands, which occurred in North Konawe and Morowali.[20] Over the next three years, deforestation will become more widespread given that in mid-June 2024 the government approved the Annual Working and Budget Plan (RKAB) for 470 Mining Business Licenses for Performing Production (IUP-OP) holders with a total quota of 240 million nickel ore per year until 2026.

Nickel processing is an energy-intensive industry and is **carbon-dense**. Compared to sulfide nickel ore, laterite ore processing consumes 3-5 times more heating energy because of the high water content in the ore.[21] RKEF process for laterite is the most energy-consuming, therefore the most carbon intensive and mainly relies on fossil fuels (coal and gas). Although producing lower emissions than RKEF, HPAL technology is still carbon-dense because the processing method requires high temperatures and pressures.[22] The HPAL process for laterite will produce 19 tons of carbon dioxide emissions per 1 ton of nickel metal (tCO2/tNi). While the RKEF procedure for NPI-to-nickel matte conversion yielded 59 tCO2/tNi. The smelting of sulfide ore is the lowest, which contains 10 tCO2/tNi.[23]

Indonesia's nickel processing industry is heavily dependent on captive coal-fired power plants as a source of energy. The government estimates that the electricity demand for nickel smelters in Sulawesi will reach 11,139 MW by 2030.[24] Currently more than 10,000 megawatts (MW) of "off-grid" coal-fired power plants are already in operation that are built in an integrated manner with nickel-based industrial parks in Indonesia. It seems that the construction of new coal-fired power plants by Chinese investors will continue despite the fact that Chinese President Xi Jinping announced in 2021 that China will stop new coal-fired power plant projects abroad.[25]

19 Ceceng Suhana. 2022. *Optimalisasi Perlindungan Hutan melalui Upaya Kolaboratif di Wilayah Kesatuan Pengelolaan Hutan Tepe Asa Maroso*. Palu: Badan Pengembangan Sumber Daya Manusia Central Sulawesi Province.

20 Erwan Hermawan. 2022. "Habis Nikel Terbitlah Deforestasi". *Tempo*, January 31-February 6.

21 Gavin M. Mudd. 2010. "Global trends and environmental issues in nickel mining: Sulfides versus laterites." *Ore Geology Reviews*, 38(1–2): S.9–S.26

22 Matthew J. Eckelman. 2010. "Facility-level energy and greenhouse gas life-cycle assessment of the global nickel industry". *Resources, Conservation and Recycling* 54:256–266.

23 IEA. 2021. "GHG emissions intensity for class 1 nickel by resource type and processing route". Available at https://www.iea.org/data-and-statistics/charts/ghg-emissions-intensity-for-class-1-nickel-by-resource-type-and-processing-route. Accessed July 22, 2024.

24 ESDM. 2024. "Penuhi Kebutuhan Listrik Smelter di Sulawesi, Pemerintah Manfaatkan Aliran Gas Donggi Senoro dan Eni Muara Bakau". *Press release*, 5 August.

25 Xinhua. 2021. "Xi Focus: China to stop building new coal-fired power projects abroad: Xi". *Xinhua,* September 22. Available at http://www.news.cn/english/2021-09/22/c_1310201218.htm. Accessed August 5, 2024.

A major concern is that the Indonesian government, through Presidential Regulation Number 112/2022 concerning the Acceleration of Renewable Energy Development for the Supply of Electricity,[26] allows the construction of new coal-fired power plants, especially those that are integrated with the construction of nickel smelters that include a transition time condition with a benefit for investors.

Nickel industry players indeed do not intend to switch to using clean energy. In recent years, they have only made words. In 2021, Tsingshan announced plans to build 2,000 MW of clean power plants at IMIP and IWIP.[27] But the announcement seems to be simply answering criticism of the dirty energy that powers the Indonesia's nickel industry. After four years, there is no real action by the company to build clean power plants.

The second gas and steam power plants with an installed capacity of 1,800 MW will also be constructed by utilizing a natural gas supply of 337 MSCF from Liquified Natural Gas (LNG) Donggi Senoro in Banggai, Central Sulawesi. Electricity from these plants will be distributed through a 500-kV extra-high voltage transmission line to nickel processing centers in South Sulawesi and Southeast Sulawesi.[28]

Tailing Transport Truck Area at IMIP. Image Source: Arianto Sangadji.

[26] Article 3, Paragraph 4 of Presidential Regulation (*Pasal* 3 *ayat* 4 *Perpres*) No. 112/2022, On the one hand, prohibiting the development of new coal-fired power plants, on the other hand, allowing as long as it is integrated with natural resource processing industry projects or classified as national strategic projects (PSN), having a commitment to reduce GHG emissions by at least 35% in 10 years since operation, and operating until 2050.

[27] Anonymous. 2021."Tsingshan targets 2 GW clean power base for Indonesia battery materials". Available at https://auto.economictimes.indiatimes.com/news/auto-components/tsingshan-targets-2-gw-clean-power-base-for-indonesia-battery-materials/81396124. Accessed August 5, 2024.

[28] ESDM. "Penuhi Kebutuhan Listrik Smelter di Sulawesi, Pemerintah Manfaatkan Aliran Gas Donggi Senoro dan Eni Muara Bakau".

Tailings. Nickel processing using HPAL technology produces large volumes of toxic tailings. It is estimated that every ton of nickel metal processed through the HPAL produces 100 tons of residue.[29] Government Regulation Number 22 of 2021 concerning the Implementation of Environmental Protection and Management classifies tailings as Toxic and Hazardous Materials Special with hazard category 2 (two) which is considered to have chronic and long-term toxicity related to the impact on humans and the environment. Therefore, tailings must be treated as Toxic and Hazardous Materials waste.[30]

After the government banned the discharge of tailings into the sea, HPAL projects in Indonesia operating the "dry-stack tailings" method as an alternative. However, the term "dry-stack" is considered to be inappropriate because the tailings that are claimed to be dry still contain between 15 percent and 20 percent of water.[31] Therefore, the application of this method in wet climates with high raindrops is considered challenging.

After being exposed to rain, the pile of tailings slurry that seems to be dry becomes wet again and undergoes oxidation and leaching of heavy metals. When rainfall flows through the tailings, heavy metals and other elements can be carried into the soil and surrounding water bodies.[32]

Tailings storage on land by using dams also has some risks in the earthquake-prone areas such as in Sulawesi and North Maluku. Heavy rainfall and earthquakes have the potential to cause dam failure. Indonesia must learn from the experience of a disaster that killed 300 people due to the collapse of two tailings dams (Brumadinho Dam and Samarco Dam) owned by Vale in Brazil.[33]

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29 Angela Durrant. 2023. "The Rise and Rise of Indonesian HPAL – Can It Continue?". Available at https://www.woodmac.com/news/opinion/rise-of-indonesian-hpal/. Accessed August 22, 2024; Cheen Aik Ang. 2017. *Green Processing and Waste Valorization: Sulfur Removal and Hematite Recovery from High Pressure Acid Leach Residue for Steelmaking.* Thesis Master of Applied Science Department of Chemical Engineering and Applied Chemistry University of Toronto.

30 Government Regulation (*Peraturan Pemerintah)* No. 22 of 2021 on the Implementation of Environmental Protection and Management.

31 See Steven H. Emerman. 2024. "A letter to Callum Russel and Sophie Grig". March 14.

32 Steven H. Emerman. 2023. "Evaluation of the Filtered Tailings Storage Facility in the Updated Proposal for the Savannah Lithium Barroso Mine, Northern Portugal." *Report prepared at the request of Associações Unidos em Defesa de Covas do Barroso e Povo e Natureza do Barroso*, Revise on April 13; Durrant, "The Rise and Rise of Indonesian HPAL – Can It Continue?"; Eri Silva. 2023. "Indonesia's nickel processing boom raises questions over tailings disposal". *S&P Global*, April 17. Available at https://www.spglobal.com/marketintelligence/en/news-insights/latest-news-headlines/indonesia-s-nickel-processing-boom-raises-questions-over-tailings-disposal-75180844. Accessed August 22, 2024.

33 5th November 2015, the collapse of the tailings dam of the Samarco iron ore processing company in Brazil which released 40 million cubic meters of tailings into settlements and rivers, killing 19 people; On January 25, 2019, there was a catastrophic incident of the collapse of the Brumadinho Dam which stored the iron tailings run by Vale in Brazil. This incident resulted in the death of 270 people, including 2 pregnant women. See Samarco.ND. "Biennial Report 2015-2016". Available at https://www.samarco.com/wp-content/uploads/2024/06/Samarco_Relatorio-Bienal-2015_2016-final-20170920_versao-ingles.pdf; Dom Phillips. 2016. "Samarco dam collapse: one year on from Brazil's worst environmental disaster". *The Guardian*, October 15. Available at https://www.theguardian.com/sustainable-business/2016/oct/15/samarco-dam-collapse-brazil-worst-environmental-disaster-bhp-billiton-vale-mining. Accessed July 20, 2024; Katy Watson. 2020. "'Vale ended our lives': Broken Brumadinho a year after

The brutal exploitation of workers is one of the dirty faces of the Indonesian nickel industry. Exploitation is characterized by low wages or low salary, long working hours, poor Occupational Health and Safety standards, and the eradication of unions. Indonesian workers and China workers who are the backbone of the rapid growth of the nickel processing industry have faced ruthless exploitation.[34] This situation is key for nickel processing companies to be able to reap higher profits.

Land grabbing is one of the fundamental_concerns throughout the historical development of the nickel industry. Overlap between mining concessions and agricultural lands and settlements has always been a major source of troubles. Land disputes between mining companies and local communities or indigenous peoples are inevitable and often lead to criminalization or intimidation of local peoples by security forces, as has happened in Morowali for more than 10 years.[35]

dam collapse." *BBC*, January 25. Available at https://www.bbc.com/news/world-latin-america-51220373. Accessed July 20, 2024.

34 See Arianto Sangadji et al. 2019. *Road to Ruin: Challenging the sustainability of nickel-based production for electric vehicle batteries*. Quezon City: Rosa Luxembur Stiftung, China Labor Watch. 2023. *Compounding Vulnerabilities: Chinese Workers in Indonesia*. New York: China Labor Watch; Sembada Bersama. 2024. *The Morowali Explosion: An Industrial Accident That Killed Workers in the Heartland of the Nickel Industry.* Jakarta: Sembada Bersama; Y. Wasi Gedepuraka Ruth Indiah Rahayu Ken Budha Kusumandaru. 2021. *Kesenjangan Tata Kelola Hak-Hak Buruh Sektor Pertambangan Mineral Logam di Indonesia.* Jakarta: Inkrispena; Khairul Anam et al. 2024. "Bersimpang Jalan Mengusut Ledakan". *Tempo*,14 January.

35 See Andika. 2014. "Booming Nikel, MP3EI, dan Pembentukan Kelas Pekerja". *Working Paper Sayogyo Institute* No.19.

SECOND SECTION
METHODOLOGY

This study focuses on the production of nickel for EV batteries using HPAL technology with an emphasis on the upstream supply chain, namely nickel mining and processing operations. The study uses a case study approach to investigate mining companies and nickel processing companies operating in IMIP or closely related to the park. The industrial park is the oldest and most integrated nickel industrial estate in Indonesia, covering an area of 3,000 hectares. These companies are PT Hengjaya Mineralindo (HM), PT Sulawesi Cahaya Mineral (SCM), PT Huayue Nickel Cobalt (HNC), and PT Qing Mei Bang New Energy Materials (QMB). HM and SCM are the nickel mining companies producing and supplying nickel limonite for processing. HNC and QMB are manufacturing MHP. All these companies are located in the IMIP or integrated within the industrial estate.

The four companies were selected because they represent a monopoly feature of nickel industry in Indonesia. HM is a subsidiary of Nickel Industries, a Sydney based company that also controls several nickel processing companies in IMIP and in IWIP, in a collaboration with Tsingshan Group. SCM is a joint venture between a leading Indonesian conglomerate, Merdeka Battery Materials Group (MBMA) and Tsingshan Group. The cooperation between the two groups also takes place through nickel processing companies inside IMIP. HNC is a subsidiary of Zhejiang Huayou Cobalt, a China's world-class business group that owns nickel processing facilities in IMIP and IWIP, and is building similar facilities in Pomalaa, Southeast Sulawesi. QMB is a subsidiary of GEM Co., Ltd., a giant Chinese producer of materials for new energy that is also developing nickel projects in Indonesia.

This study uses data based on: (1) the field interviews. The interviewees were informed and aware of the purpose of this study. To protect workers from potential threats and intimidation from the companies they have worked for, the study quotes their opinions anonymously or pseudonymously. The same treatment is also given to the villagers who participated in interviews; (2) a review of official documents or reports produced by the companies, relevant research, and news and investigative reports from the mainstream media. The data collected during the study was then analyzed and presented qualitatively and quantitatively.

IMIP. Image Source: indonesiabusinesspost.com

THIRD SECTION
INDONESIA MOROWALI INDUSTRIAL PARK:
THE NICKEL PRODUCTION SUPPLY CHAIN FOR EV BATTERIES

1. Mining

Most of the supply of low-grade nickel ore for the HPAL projects at IMIP sources from mining operations in Southeast Sulawesi and Central Sulawesi. From both regions, most of it comes from mining companies in Routa in Konawe Regency (see **Table 1**) and in Morowali Regency (see **Table 2**). These companies operate in locations relatively close to IMIP, making ore transportation cost efficient and constantly supply limonite ore to the park. Compared to other nickel mining companies in the regions, HM and SCM are two of the few mining companies affiliated with the Tsingshan Group, the Chinese giant that controls IMIP.

Apart from Central Sulawesi and Southeast Sulawesi, limonite supplies arriving at Morowali Port also come from the Philippines since 2023. Indonesia imported more than 374,000 tons of nickel ore from the Philippines worth USD 16 million last year, and about 500,000 tons of ore in April 2024. This is due to delays in receiving the Annual Working and Budget Plan approval from the Ministry of Energy and Mineral Resources.[36]

Nickel Ore.
Image Source: millops.community.uaf.edu

[36] Mai Nguyen & Siyi Liu. 2023. "Indonesian nickel smelters turn to Philippines for ore as local supply tightens". August 30. Available at https://www.reuters.com/markets/commodities/indonesian-nickel-smelters-turn-philippines-ore-local-supply-tightens-2023-08-30/. Accessed July 25, 2024.BPS. 2024. *Indonesia Foreign Trade Statistics 2023 I Import Book*. Jakarta: BPS; Anonymous. 2024. "Indonesia buys record amounts of Philippine nickel ore due to quota delays". *Mining com*, May 29. Available at https://www.mining.com/web/indonesia-buying-record-amounts-of-philippine-nickel-ore-due-to-quota-delays-sources-say/. Accessed July 25, 2024.

Table 1. Nickel Mining Business License Area for Performing Production in Routa District, Konawe Regency, Southeast Sulawesi

Nama perusahaan	Luas (ha)	Masa berlaku
Sulawesi Cahaya Mineral	21,100.00	2019-2037
Modern Cahaya Makmur	5,053.00	2012-2032
Intan Perdhana Puspa	3,000.00	2013-2033
Pelangi Utama Jaya Mandiri	2,375.00	2024-2032
Nikelindo Suryakencana Agung	2,654.00	2017-2032
Alvindo Mining Resources	2,439.00	2012-2032
Andalan Energi Nusantara	403.20	2017-2032
Putri Glory Elshaday	4,086.00	2023-2032
Berkah Taluri Jaya	1,130.00	2024-2032
Homkey Inti Prima	4,870.00	2024-2032

Source: Processed from Momi ESDM (2024)

Table 2. Nickel Mining Business License Area for Performing Production in Morowali Regency

Company	Area (Ha)	Validity Period	Location (Village)
Ang and Fang Brother	775.00	2022-2032	Lalampu (Bahudopi)
Bima Cakra Perkasa Mineralindo	199.00	2013-2030	Laroenai (Bungku Pesisir)
Bineka Karya	970.00	2022-2032	Bahudopi
Bintang Delapan Mineral	20,765.00	2022-2027	Bahomoahi, Bahomotefe, Lalampu, Lele, Dampala, Siumbatu, Fatufia, Bahodopi, Keurea
Bintang Sinar Perkasa	1,867.70	2017-2036	Lamontoli (Bungku Selatan)
Bumi Kalaena Persada	685.00	2011-2031	Bahodopi (Bahudopi)
Bumi Morowali Utama	1,963.00	2011-2026	Laroenai
Bratama Mercindo Abadi	1,199.00	2011-2031	Laroenai
Cahaya Ginda Ginda	1,942.00	2010-2030	Siumbatu, Bahudopi
Cetara Bangun Persada	199.00	2021-2031	Lalampu
Erabaru Timur Lestari	1,159.00	2011-2031	Lele, Dampala, Siumbatu
Fadlan Mulia Jaya	199.00	2021-2031	-
Gita Flora	624.00	2024-2030	Dampala
Graha Mining Utama	1,102.00	2022-2032	Bahodopi
Graha Mining Utama	624.53	2012-2032	Lalampu

Company	Area (Ha)	Validity Period	Location (Village)
Hengjaya Mineralindo	5,983.00	2020-2031	Padabaho, Bete Bete, Puungkeu, Tangofa
Indoberkah Jaya Mandiri	196.00	2014-2034	Torete
Kencana Bumi Meneral	2,549.00	2018-2028	Lele, Dampala, Lalampu, Bahodopi
Laroenai Bungsel Sorajai	256.00	2018-2026	-
Mahligai Artha Sejahtera	514.00	2021-2031	Bungku Pesisir
Mandiri Jaya Nikel	4,871.00	2014-2034	Bahudopi
Marco Puri Indah Perkasa	3,480.00	2011-2031	-
Mitra Jaya Perkasa Abadi	1,086.00	2024-2032	Bahudopi & Bungku Pesisir
Mulai Dari Indonesia	3,147.00	2011-2031	Bahudopi & Bungku Pesisir
Nusajaya Persadatama Mandiri	1,852.00	2012-2025	Matarape
Nusa Mineral Semesta	1,015.00	2023-2031	Dampala
Oti Eya Abadi	3,339.00	2018-2038	Lele, Dampala, Siumbatu
PAM Mineral	198.00	2012-2025	Buleleng, Laroenai
Putra Sulawesi Mining	650.00	2013-2033	Torete, Lafeu
Raihan Catur Putra	688.00	2013-2035	Torete
Sarana Maju Cemerlang	538.00	2022-2032	Bahodopi
Selaras Maju	572.00	2014-2034	Lalampu
Sumber Swarna Pratama	4,023.00	2014-2034	Matano, Matarape
Teratai Bindo Utama	319.00	2012-2032	Bahomotefe
Teknik Service	1,301.00	2012-2029	Buleleng, Torete
Tiga Baji	199.00	2022-2029	Matarape
Total Prima Indonesia	1,142.00	2011-2031	Tangofa
Wosindo Perkasa	1,825.00	2021-2031	Sambalagi
Vale Indonesia	22,699.00	2024-2035	Bahudopi

Source: Processed from MOMI ESDM (2024)

1.1. PT Hengjaya Mineralindo - Nickel Industries

HM is a foreign direct investment company in which Nickel Industries Limited (ASX: NIC), a Sydney-based multinational company controls an 80 percent stake in HM. The Indonesian Wijoyo family controls the remaining a 20 percent.[37] One of Wijoyo family members Adi Wijoyo also holds a 75 percent stake in PT Erabaru Timur Lestari (ETL), which controls 1,159 hectares of Mining Business License located in Dampala Village, Siumbatu Village and Lele Village in Morowali Regency and a 10 percent stakes in PT Genba Multi Mineral, which has a Mining Business License of 3,744 hectares in Molino Village and Bimor Jaya Village in North Morowali Regency.

In addition to owning HM, Nickel Industries also holds a majority stake in several nickel mining companies in Morowali. This year, Nickel Industries acquired over 60 percent of shares in each of the three nickel mining companies within the "Sampala Project", located about 40 kilometers from IMIP. The three nickel companies include PT Erabaru Timur Lestari, PT Mandiri Jaya Nikel (MJN), and CV Gita Flora (GF) that have total resources of approximately 2.3 million tons of nickel metal and 200,000 tons of cobalt metal. Nickel Industries stated that with the acquisition, the combined resources of the Sampala and HM Projects will be economically viable for between 40 and 50 years.[38]

Nickel Industries in collaboration with Tsingshan Group also has several nickel processing companies. Among these are PT Hengjaya Nickel Industry (HNI), PT Ranger Nickel Industry (RNI), and PT Oracle Nickel Industry (ONI) at IMIP as well as PT Angel Nickel Industry (ANI) at IWIP. Nickel Industries also take part in two HPAL projects, i.e. PT Huayue Nickel Cobalt (HNC) and PT Excelsior Nickel Cobalt (ENC) at IMIP (see **Figure 4**).

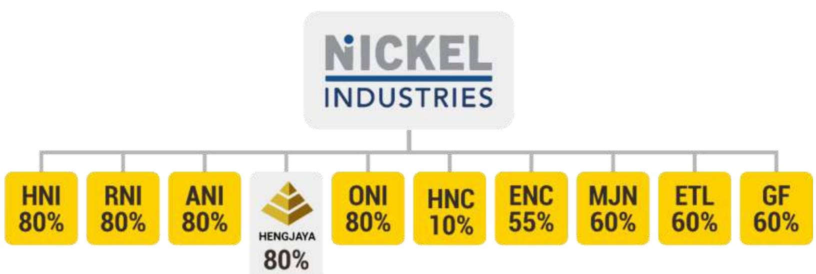

Figure 4. Nickel Industries Business Arm in Indonesia

[37] Nickel Industries. 2024. *2023 Annual Report*. Sydney: Nickel Industries.

[38] Nickel Industries. 2024. "Aqqusition of World Class Nickel Portfolio". Available at https://cdn-api.markitdigital.com/apiman-gateway/ASX/asx-research/1.0/file/2924-02853750-2A1548964. Accessed September 25, 2024.

As a public company listed on the Australian Securities Exchange (ASX) Sydney, Nickel Industries shares are held by several foreign companies, including Tsingshan Group which controls about a 21 percent stake through its subsidiaries. Several Indonesian mining companies such as PT Danusa Tambang Nusantara, a subsidiary of PT United Tractors Tbk (UNTR), and PT Harum Energy Tbk (HRUM) also hold stakes in Nickel Industries. The company's shares are also controlled by some financial institutions affiliated with global giants such as HSBC, Citicorp, JP Morgan and BNP Paribas (see **Table 3**).

Given it is the oldest subsidiary of Nickel Industry in Indonesia, HM has had a nickel mine concession in Morowali Regency since the beginning of the last decade. The Regent of Morowali issued the Mining Business License with an area of 6,249 hectares in 2011. The Governor of Central Sulawesi reduced the area to 5,983 hectares in 2020. The area includes both sides of the Trans Sulawesi that connects Bete-Bete village in Bahudopi sub-district with Tangofa village in Bungku Pesisir sub-district. HM's Mining Business License is valid from 2011 to 2031 and can be extended twice for each ten years.[39]

Table 3. Nickel Industries Limited: Major Shareholders (2023)

Company	Shares (%)
PT Danusa Tambang Nusantara	19.98
Decent Investment International Private Limited	15.68
HSBC Custody Nominees Limited	13.26
Citicorp Nominees Pty Limited	9.56
JP Morgan Nominees Australia Pty Limited	6.25
PT Harum Energy	4.16
Shanghai Decent Investment (Group) Co Ltd	3.77
BNP Paribas Noms Pty Ltd	3.72
Decent Resource Limited	2.52
Shanghai Wanlu Investment Co Ltd	2.27

Source: Processed from Nickel Industries (2024)

[39] Regent Decree (*SK Bupati*) of Morowali No. 540.K/SK.001/DESDM/VI/2011 on the Approval of the Upgrade of Exploration Mining Business License to Production Operation Mining Business License for PT Hengjaya Mineralindo and Governor Decree (*SK Gubernur*) of Central Sulawesi No. 540/345/IUP-OP-Penciutan/DPMPTSP/2020 on the Reduction of the Mining Business Area for Production Operation of PT Hengjaya Mineralindo Based on the Regent Decree (*SK Bupati*) of Morowali No. 540.K/SK.001/DESDM/VI/2011 on the Approval of the Upgrade of Exploration Mining Business License to Production Operation Mining Business License for PT Hengjaya Mineralindo.; Also see MODI ESDM. 2024. PT Hengjaya Mineralindo. Available at https://modi.esdm.go.id/portal/detailPerusahaan/3349?jp=1. Accessed August 22, 2024.

HM has mineral resources estimated at 300 million wet tons with an average grade of 1.22 percent nickel and 0.09 percent cobalt. The resource is equivalent to 3.7 million tons of nickel metal and 270 thousand tons of cobalt. Saprolite ore grading 1.8 percent is estimated at 72 million wmt) while Limonite resources are estimated at 151 million wet tons for 1.2 percent nickel and 0.14 percent cobalt. With a mine site located about 12 kilometers from IMIP, HM supplies ore to the industrial parl with lower cost transportation.[40]

HM mined 430,000 tons of nickel ore during 2012 - 2013, the period before the government banned nickel ore exports in January 2014. Most nickel ores were exported to China, with some to Japan. Since September 2015, HM's mine production has increased year-on-year basis (see **Figure 5**) initially to supply saprolite ore to NPI producers in the IMIP. HM signed nickel ore supply agreements with PT Sulawesi Mining Investment and later PT Indonesia Tsingshan Stainless Steel (ITSS).[41] In 2019, Hengjaya started supplying saprolite ore to smelting plants controlled by Nickel Industries in IMIP.

Currently, HM is one of the major suppliers of limonite ore for the nickel refining plants at IMIP. In December 2018, following the planned construction of a nickel processing plant for EV battery raw materials, the company made a memorandum of understanding with IMIP to supply nickel ore to the industrial park. In line with the operation of HPAL processing plants at IMIP since 2022, the volume of limonite ore mined by HM has increased sharply in the past 3 years. From 1.1 million bank cubic meters (BCM) in 2021, it increased to 3.9 million tons in 2022 and skyrocketed to 9.5 million tons in 2023. HM supplies the limonite to QMB and HNC. Since November 2021, HM began trucking limonite to the HNC plant, an HPAL project in which Nickel Industries holds a 10 percent stake.[42]

Claiming to be "*a global leader in low-cost nickel production*", Nickel Industries' subsidiaries are profitable companies. This includes HM, whose profits have surged over the past five years, jumping from USD 1.3 million in 2019 to USD 52.4 million in 2023 (see **Figure 5**).

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[40] Nickel Industries Limited, *2023 Annual Report.*

[41] Nickel Mines Limited. 2019. *Annual Report 2018*. Sydney: Nickel Mines Limited.

[42] Nickel Industries. 2023. *Annual Report 2022*. Sydney: Nickel Industries.

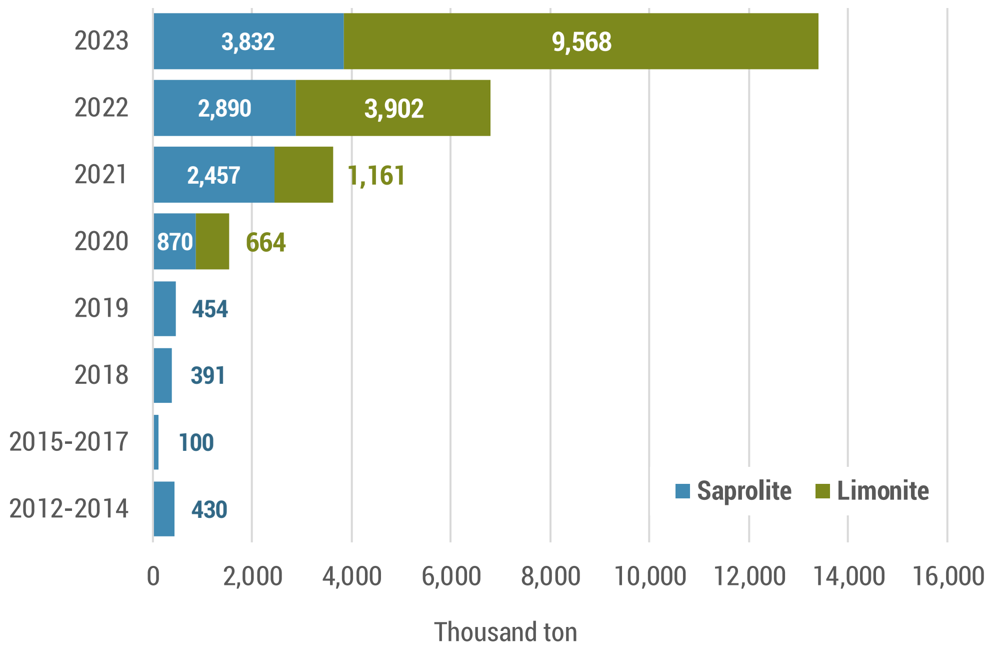

Figure 5. Hengjaya Mineralindo: Saprolite and Limonite Production, 2012-2023 (in Thousand Tons)

Source: Processed from HM data

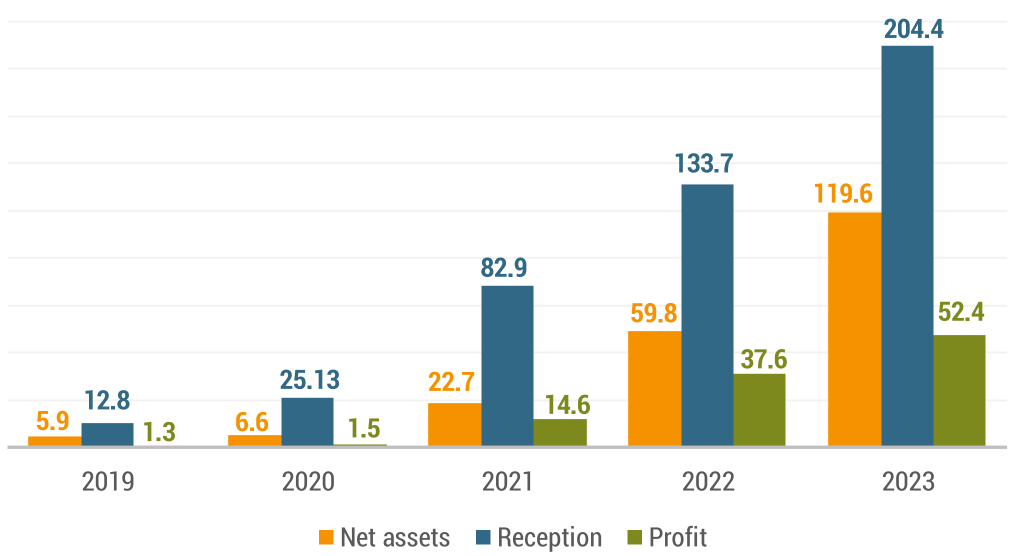

Figure 6. Hengjaya Mineralindo: Financial Performance 2019-2023 (in Million USD)

Source: Processed from Nickel Industries annual report

1.2. PT. Sulawesi Cahaya Mineral – Merdeka Group

SCM is part of a joint venture between PT Merdeka Battery Materials Tbk (MBMA) and Tsingshan Group. MBMA, through its subsidiary PT Merdeka Industri Mineral (MIN), holds a 51 percent stake in SCM, while Tsingshan Group owns the remaining 49 percent.[43]

MBMA is one of the largest national business groups and has steadily involved in integrated nickel projects in Indonesia. PT Merdeka Copper Gold Tbk (MDKA), through PT Merdeka Energi Nusantara (MEN), holds a 50.04 percent stake in MBMA. In addition to MEN, MBMA shares are held by Garibaldi Thohir (8.46%), Huayong International (Hongkong) Limited (7.55%), PT Alam Permai (5.43%), Contemporary Brunp Lygend Co.Ltd., a subsidiary of CATL (5%), with the remaining 28.52% held by shareholders with less than 5% ownership.[44] PT Saratoga Investama Sedaya Tbk is an 18 percent stake in Merdeka Copper Gold, Garibaldi Thohir holds a 7.46 percent stake, and Sakti Wahyu Trenggono (former Deputy Minister of Defense in Jokowi's administration and Minister of Marine Affairs and Fisheries under President Jokowi and President Prabowo) holds a 0.99 percent stake. Garibaldi Thohir is the older brother of Erick Thohir (Minister of State-Owned Enterprises under Jokowi and Prabowo). Sandiaga Salahuddin Uno (Minister of Tourism and Creative Economy under Jokowi) is a 21.51 percent stake in Saratoga Investama Sedaya.[45]

Tsingshan Group, a Chinese nickel giant (known as the "Nickel King"), controls various nickel and stainless steel processing facilities in Indonesia. With a strong presence in IMIP and IWIP, Tsingshan has pioneered Rotary Kiln Electric Furnace (RKEF) projects, stainless steel production, and industrial parks in Indonesia since the mid-2010s.[46]

SCM is just one part of the integrated nickel mining and processing business network between MBMA and Tsingshan Group. In addition to controlling SCM, MBMA through PT Merdeka Industri Mineral (MIN) controls a 50.10 percent stake in each of PT Cahaya Smelter Indonesia (CSI), PT Bukit Smelter Indonesia (BSI), and PT Zhao Hui Nickel (ZHN), which produce NPI. The remaining equities of the companies operating nickel smelters in IMIP are held by Tsingshan Group. MBMA through PT Merdeka Mega Industri (MMI) also controls a NPI-to-nickel matte conversion project operated by PT Huaneng Metal Industry (HNMI). MBMA holds a 60 percent stake and Tsingshan owns a 40 percent stake in HNMI. Finally, MBMA through PT Batutua Pelita Investama controls 80 percent and Tsingshan Group holds 20 percent stake in PT Merdeka Tsingshan Indonesia (MTI) which produces sulfuric acid at IMIP.[47]

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43 PT Merdeka Battery Materials Tbk. 2024. *2023 Annual Report.* Jakarta: PT Merdeka Battery Materials.

44 *Ibid*.

45 *Ibid*.

46 Arianto Sangadji. 2020. "Tata Kelola Sumber Daya Alam di Sulawesi Tengah: Pengalaman Industri Berbasis Nikel di Morowali." Auriga and KPK *Working Paper*; Arianto Sangadji & Pius Ginting, 2023. "Peran Perusahaan Multinasional dalam Hilirisasi Nikel di Indonesia." Jakarta: AEER

47 *Ibid*.

Meanwhile, through PT Merdeka Energi Industri (MED), MBMA also holds a 68 percent stake in Indonesia Konawe Industrial Park (IKIP), a new industrial park located in SCM's nickel site in Routa, Konawe Regency in Southeast Sulawesi. Tsingshan Group holds a 32 percent stake in the 3,500-hectare industrial park.

In collaboration with Chinese giants, MBMA is also entering hydrometallurgy projects. The Merdeka Group has established a partnership with GEM Co., Ltd to build a HPAL plant at IMIP with a production capacity of 30,000 t/y of MHP. The HPAL project is operated by PT ESG New Energy Material (ESG). MBMA, through PT Merdeka Industri Anantha (MIA) and GEM control 60% and 40% respectively of ESG's stakes.[48] MBMA is also partnering with Contemporary Brunp Lygend Co. Ltd - a subsidiary of CATL - to develop a HPAL plant with a capacity of 60,000 t/y of MHP. In this project MBMA holds a 67 percent stake and Brunp holds the remaining 33 percent stake. Brunp has a 5 percent stake shareholder in PT Merdeka Copper Gold Tbk, MBMA's parent company.[49] The HPAL plant and tailings processing facility at IKIP are expected to be operational in 2026 or 2027.[50]

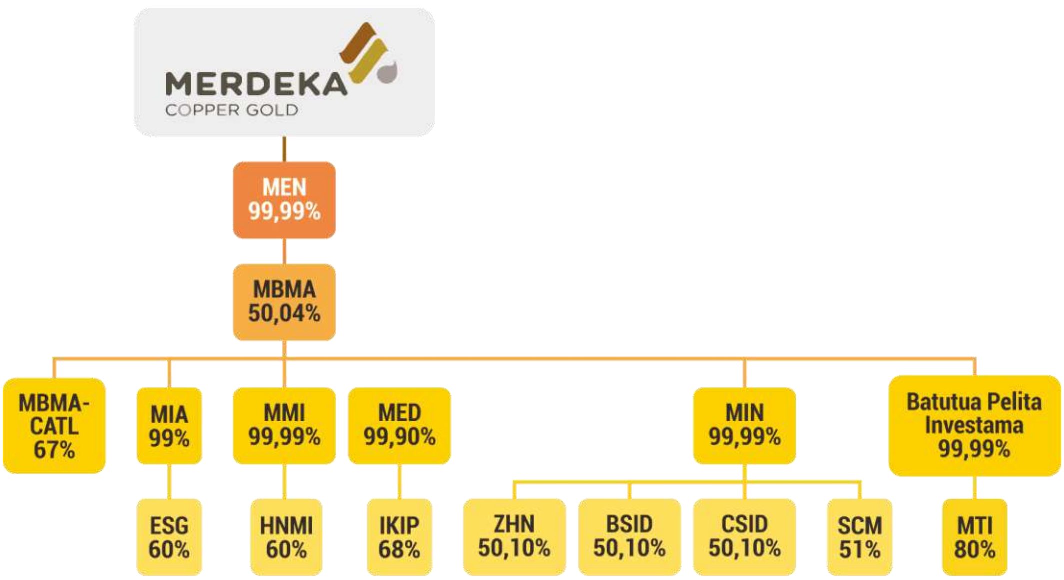

Figure 7. SCM diagram of Merdeka Copper Gold Group's business octopus

48 *Ibid*.

49 PT Merdeka Copper Gold Tbk. 2024. *2023 Annual Report*. Jakarta: Merdeka Copper Gold Tbk.

50 PT Merdeka Copper Gold Tbk. 2024. *Merdeka Copper Gold (Presentation)*. Available at https://merdekacoppergold.com/wp-content/uploads/2024/06/MDKA-Investor-Presentation-June-2024.pdf Accessed August 24, 2024

SCM holds a Mining Business License Area for Performing Production in an area covering 21,100 hectares in Routa, Konawe and is valid until 2037.[51] In addition to SCM, there are nine holders of Mining Business Licenses for Performing Production operating in Routa, a sub-district with an area of nearly 219,000 hectares. The total area of Mining Business Licenses for Performing Production in the sub-district is 47,010.2 hectares, or about 21.48 percent of the sub-district area.

SCM was originally owned by global mining giant Rio Tinto. Rio Tinto had been exploring nickel in the region since the mid-2000s. At that time, Rio Tinto held a nickel concession of 95,130 hectares, known as the Lasampala Block, covering areas inside three provinces (Central Sulawesi, Southeast Sulawesi and South Sulawesi), before it was reduced to 44,067 hectares. Before withdrawing from the project in the late 2000s, Rio Tinto had planned to build a US 1 billion nickel processing facility with a production capacity of 45,000 t/y. MBMA and the Tsingshan Group then control SCM.[52]

SCM's mining concession area is one of the largest nickel deposits in the world. The company controls reserves of 1.1 billion dmt of ore with nickel grade of 1.22 percent and cobalt grade of 0.08 percent containing 13.8 million tons of nickel metal and 1 million tons of cobalt metal. Limonite resources account for 70 percent and saprolite 23 percent.[53]

SCM supplies nickel ore to the nickel processing plants at IMIP. The distance between SCM's mine site and IMIP is about 50 kilometers. The company delivers saprolite ore to RKEF smelters operated by CSI, BSI, and ZHN that are also Merdeka-Tsingshan joint projects. It supplies limonite ore to HNC's HPAL plant in IMIP. For HPAL projects, a major target of SCM is to supply limonite that can be converted into 300,000 t/y of MHP. By 2024, SCM expects to extract between 10 and 11 million tons of limonite.[54]

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51 Modi.2024. "Sulawesi Cahaya Mineral." Available at https://modi.esdm.go.id/portal/detailPerusahaan/7057?jp=1. Accessed September 20, 2024; PT Merdeka Battery Materials Tbk. 2023. *Annual Report 2022*. Jakarta: PT Merdeka Battry Materials Tbk.

52 Rio Tinto. 2009. *Annual Report 2008*. London: Rio Tinto; see also Andika, "Booming Nikel, MP3EI, dan Pembentukan Kelas Pekerja"; Anonymous. 2007. "Rio Tinto wants Indonesia nickel mine under old law". *Reuters,* August 10. Available at https://www.reuters.com/article/world/rio-tinto-wants-indonesia-nickel-mine-under-old-law-idUSJAK153594/. Accessed August 14, 2024.

53 PT Merdeka Battery Materials Tbk, *2022 Annuarl Report*.

54 PT Merdeka Battery Materials Tbk. 2024. *Quarterly Report: March 2024.* Jakarta: PT Merdeka Battery Materials Tbk, 2023; PT Merdeka Battery Materials Tbk. 2024. *Quarterly Report: June 2024.* Jakarta: PT Merdeka Battery Materials Tbk, 2023.

Huayou Cobalt holds a 7.55 percent stake in MBMA that ensures SCM securely supplies limonite to HNC. As of the second quarter of 2024, SCM has delivered 3.96 million tons of limonite to HNC. Since the beginning of 2024, SCM began transporting limonite ore from the stockpile to HNC's feed preparation plant (FPP) in SCM's concession area. By building a slurry pipeline that extends from HNC's FPP to IMIP for 62 kilometers, limonite delivery goes through pipes after nickel is mixed with water into slurry. This slurry delivery will also be used for the HPAL facility of PT ESG New Energy Material (ESG) which is a joint venture of MBMA and GEM Co., Ltd. Under a 20-year limonite delivery agreement, SCM will supply nickel via pipeline to ESG's plant at IMIP.[55]

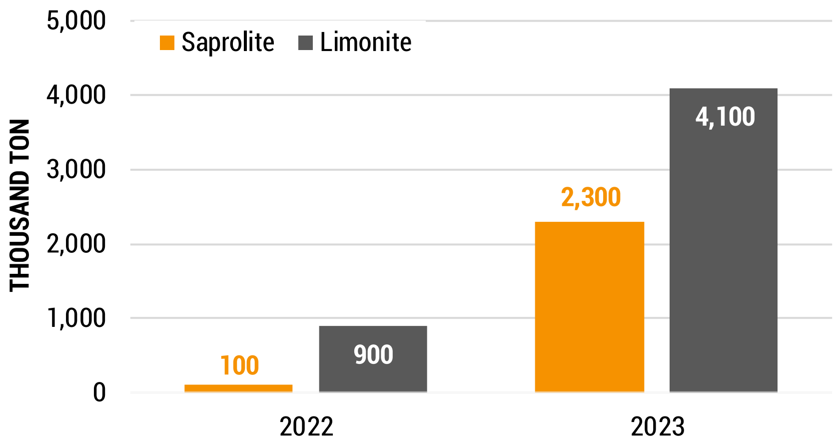

Figure 8. Sulawesi Cahaya Mineral: Nickel Production 2022-2023 (in Thousand Tons)

Source: Processed from MBMA data

Limonite & Saprolite.
Image Source:
Jordan / ftmmachinery.com

[55] PT Merdeka Battery Materials Tbk, *2023 Annual Report*; Huayou Cobalt, *2023 Annual Report*. Zhejiang: Huayou Cobalt; Huayou Cobalt. 2024. *2023 Environmental, Social and Governance (ESG) Report.* Zhejiang: Huayou Cobalt.

2. Nickel Refining

Investment in nickel processing for EV batteries using HPAL technology has grown rapidly since the beginning of this decade. Five existing HPAL projects are operating across the archipelago. These HPAL facilities have a production capacity of approximately 355,000 tons of MHP per year (see **Figure 9**). HNC and QMB are two of the five operating projects.

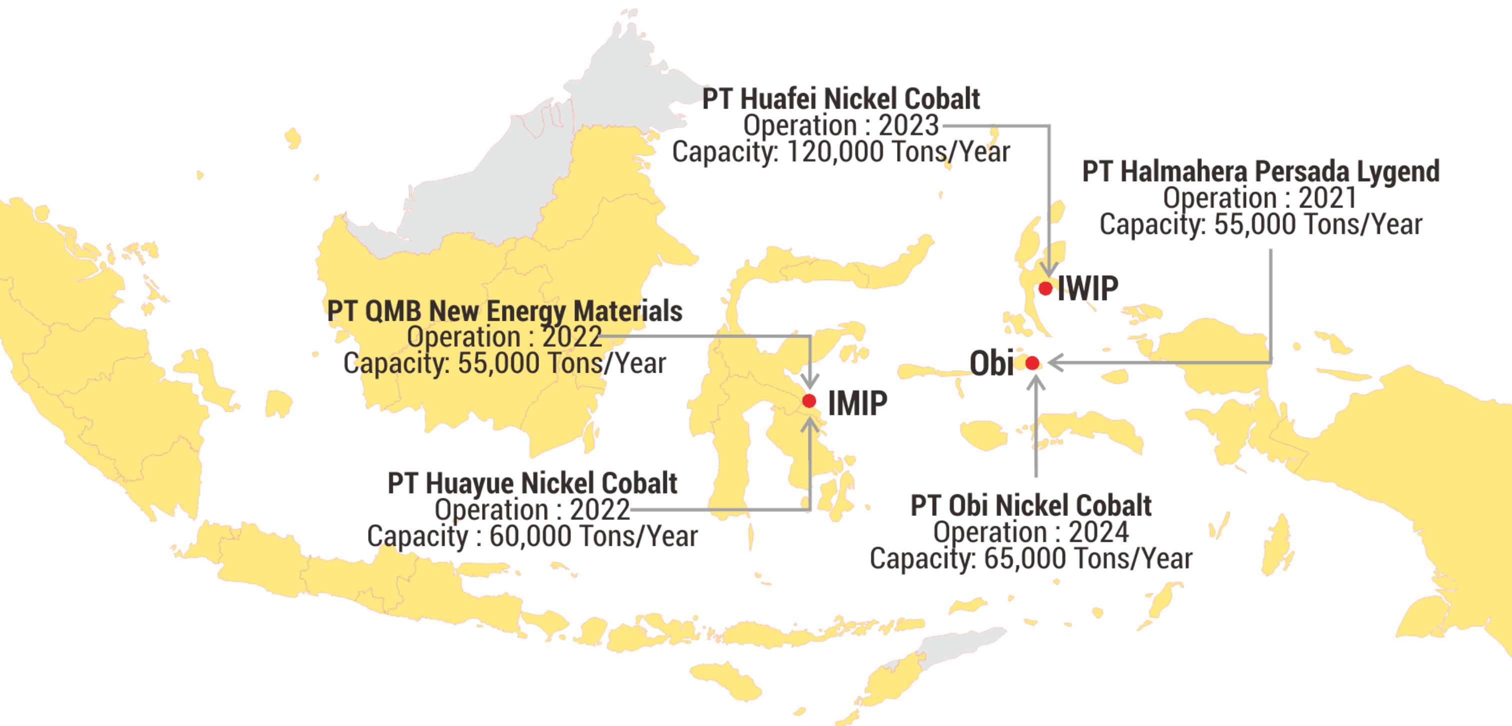

Figure 9. Existing HPAL Projects in Indonesia

2.1. PT Huayue Nickel Cobalt - Zhejiang Huayou Cobalt Co. Ltd

Huayue Nickel Cobalt (HNC) is a joint venture of Zhejiang Huayou Cobalt Co., Ltd. (57%), CMOC (30%), Nickel Industries (10%), Huaqing Hualong Consulting (2%), and Long Sincere (1%). HNC is the first HPAL project controlled by Huayou in Indonesia and is part of a business arm of Zhejiang Huayou Cobalt Co. Ltd (浙江华友钴业股份有限公司), a Chinese metal industry giant that has heavily invested in the nickel-based new energy industry in the country (see **Figure 10**).

Table 4. Huayou Cobalt: Top 10 Shareholders (2023)

Investor	% Share
Huayou Holding Group Co., Ltd.	15.22
Chen Xuehua	6.43
Hangzhou Youyou Enterprise Management Partnership (Limited Partnership)	4.38
Hong Kong Securities Clearing Co., Ltd.	4.15
China Construction Bank Corporation - Huaxia Energy Innovation Equity Securities Investment Fund	1.66
Citibank, National Association	1.45
Industrial and Commercial Bank of China	0.97
China Postal Savings Bank Co., Ltd.	0.67
CITIC Securities Company Limited	0.62
Agricultural Bank of China	0.62

Source: Huayou Cobalt[56]

Huayou Cobalt is controlled by a variety of shareholders including Huayou Holding Group Co., Ltd. which holds 15.22 percent stake in Huayou Cobalt. Chen Xuehua, the founder of Huayou and one of China's superrich, holds 6.43 percent of Huayou Cobalt and indirectly owns Huayou Holding (see **Table 4**). Trading on the Shanghai Stock Exchange (stock code: 603799) and on the SIX Swiss Exchange (stock code: HUAYO), the public holds 70.35 percent of Huayou Cobalt.[57]

[56] Huayou Cobalt, *2023 Annual Report of Zheziang Huayou Cobalt Co.Ltd.*

[57] *Ibid.*

Given its scale investment in nickel projects for new energy, the total production capacity of Huayou's HPAL production lines operating in Indonesia is expected to reach over 400,000 t/y of MHP in the next few years. Of the two HPAL facilities already in operation, HNC has an installed capacity of 60,000 t/y of MHP and PT Huafei Nickel Cobalt has a production capacity of 120,000 t/y of MHP. Huafei is the largest HPAL project in the world in which Huayou Cobalt controls a 51 percent stake in the project. The remaining shares are held by Eve Battery (17%), Glaucous International (30%), and Lindo Investment (2%).[58]

Huayou also has HPAL projects under construction. At IWIP, through PT Huashan Nickel Cobalt, Huayou is building a hydrometallurgical project with an investment value of USD 2.6 billion with a production capacity of 120,000 t/y of MHP. The project is a joint venture between Huayou (68%) and Glaucous (32%).[59] Other HPAL project related to Huayou is through PT Kolaka Nickel Indonesia (KNI) located in Pomalaa Southeast Sulawesi which is a joint venture of Huayou Cobalt (73.2%), Vale Indonesia (18.3%), and Ford Motor Company (8.5%).[60] This project with an investment value of USD 4.29 billion is targeted to produce 120,000 t/y of nickel and 15,000 t/y of cobalt.

Huayou also has a definitive agreement with Vale Indonesia to build a HPAL project in Malili, East Luwu regency, South Sulawesi. Operated by PT Huali Nickel Indonesia with an investment value of USD 1.8 billion, the project is expected to produce 60,000 t/y of MHP [61].

In IWIP, Huayou has other nickel-based projects for new energy. Through its subsidiary PT Huake Nickel Indonesia, Huayou tuns a RKEF smelter with a production capacity of 45,000 t/y of nickel matte. Huayou also plans to build a nickel sulfate project with a production capacity of 50,000 t/y. PT Huaxiang Refining Indonesia, which will operate the project, is a collaboration between Huayou through Huayao International Investment Pte. Ltd. (Huayao), Strive Investment Capital Pte. Ltd. (Strive), and Lindo Investment Pte. Ltd. (Lindo). Huayou and Strive each hold a 49 percent stake and Lindo 2 percent.[62]

58 *Ibid*.

59 SMM. 2022. "Huayou Cobalt Plans to Invest in the Construction of Huashan Nickel-Cobalt through Its Wholly-Owned Subsidiary". *SMM*, June 21. Available at https://news.metal.com/newscontent/101867749/huayou-cobalt-plans-to-invest-in-the-construction-of-huashan-nickel-cobalt-through-its-wholly-owned-subsidiary. Accessed August 22, 2024; Tang Shihua. 2022. Huayou Cobalt to Raise Up to USD2.6 Billion for Indonesia Nickel Project.Yicai, June 20. Available at https://www.yicaiglobal.com/news/huayou-cobalt-to-raise-up-to-usd26-billion-for-indonesia-nickel-project Accessed August 22, 2024.

60 PT Vale Indonesia, 2023. "PT Vale Indonesia and Huayou Sign Nickel Agreement with Ford Motor Co. Supporting Growth of the Global Sustainable EV Industry". *Press Release*, March 30,

61 PT Vale Indonesia Tbk. 2024. *Annual Report 2023*. Jakarta: PT Vale Indonesia; Huayou Cobalt, *2023 Annual Report of Zheziang Huayou Cobalt Co.Ltd*.

62 Huayou Cobalt, *2023 Annual Report of Zheziang Huayou Cobalt Co.Ltd.*; Anonymous. 2023. "Chinese high-tech enterprise Huayou Cobalt to invest in 50,000t/yr nickel sulfate project in Indonesia." July 26. Available at https://en.imsilkroad.com/p/335269.html. Accessed August 22, 2024.

Huayou is also actively developing nickel-based industrial parks. IWIP is collaboratively managed by Huayou, Tsingshan Group, and Zhenshi Holding Group. Huayou is building a new industrial park called the Indonesia Pomalaa Industrial Park (IPIP) in Pomalaa, an industrial park with an integrated nickel supply chain for new energy. Another industrial estate is the Indonesia Huali Industrial Park (IHIP) in Malili, where Huali will operate a HPAL production facility.[63]

To ensure the supply security of the nickel ores, Huayou has also involved in nickel mining operations. Huayou holds a 35 percent stake in PT Anugerah Jaya Buana (AJB) and PT Wana Kencana Mineral (WKM) respectively. By owning a stake in AJB, which has a Mining Business License covering 2,800 hectares in East Luwu Regency, Huayou will secure nickel ore supply to Huali's HPAL project in Malili. Meanwhile, by owning a stake in WKM, which holds a Mining Business License covering 24,700 hectares in East Halmahera Regency and Central Halmahera Regency, Huayou has secured the supply of nickel ore for the smelters under its control in IWIP. In addition, by holding a minority stake in MBMA, the business group controlling SCM, Huayou ensures to source nickel ore for Huayou's projects at IMIP.

Figure 10. Huayou Cobalt and its Associate Companies in Indonesia

63 Huayou Cobalt, *2023 Annuar Report of Zheziang Huayou Cobalt Co.Ltd*

Huayou, through HNC, had planned to build a HPAL plant at IMIP since 2018. The project experienced delays mainly due to the coronavirus pandemic. With an investment value of USD 1.28 billion, HNC started construction of the plant in 2020. To secure investment, the Indonesian government offered tax incentives including a 10-year tax holiday for corporate income tax beginning with the start of production. After the 10-year operating period, the income tax relief is extended for 2 years at a rate of 50 percent[64].

After a trial period, the processing facility started commercial operation in January 2022 and became the first HPAL project to operate at IMIP. With 4 autoclaves, HNC has an ultimate planned capacity of 60,000 t/y of MHP. In 2023, HNC achieved a record MHP production of 70,000 tons exceeding installed capacity.[65]

HNC sources limonite from HM and SCM. HNC began to obtain about 98,300 tons of limonite from HM since trial production run in late 2021.[66] To ensure a stable supply of nickel ore, Huayou Cobalt - the parent of HNC – has indirectly invested in SCM by holding bonds of MBMA, the parent company of SCM.[67] Ore supply from HM is also secure as Nickel Industries - HM's parent company - has a 10 percent equity interest in HNC.

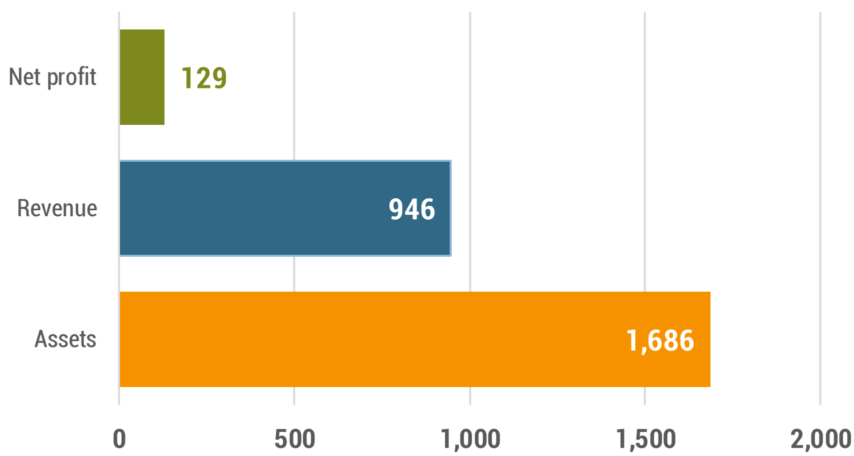

Figure 11. Huayue Nickel Cobalt: Financial Performance in 2023 (in Million USD)

Source: Extracted from Huayou[68]

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64 *Ibid.*

65 *Ibid*

66 Huayou Cobalt, *Annual Report 2022*.

67 Huayou Cobalt.2024. *2023 Annual Report*.

68 *Ibid*.

Figure 12. Huayue Nickel Cobalt Product: Mixed Hydroxide Precipitate

Source: Personal Documentation

2.2. PT Qing Mei Bang New Energy Materials – GEM Co., Ltd

Qing Mei Bang New Energy Materials (QMB) is a joint venture company in which Green-Eco-Manufacture Co., Ltd (GEM), 格林美股份有限公司, controls a 63 percent stake. As of 2024, the remaining stakes are held by IMIP (10%), Brunp Recycling Technology - a subsidiary of Contemporary Amperex Technology Co., Limited - (10%), South Korean company EcoPro Global Co., Ltd. (9%), and Japanese company Hanwa Industrial Co Ltd (8%). GEM, which leads the QMB project, established in 2001, produces NCM and NCA precursors with high nickel content for EV batteries, in addition to being known as a recycled battery manufacturer.[69]

Table 5. GEM Co., Ltd: 10 major shareholders

Investor	% Shares
Shenzhen Huifengyuan Investment Co., Ltd	8.40
HKSCC Nominees Limited	3.35
Agricultural Bank of China Limited–CSI 500 Exchange Traded Fund	1.76
Beijing Jingneng Tongxin Investment Management Co., Ltd.– Beijing Jingneng Energy	0.93
China State-owned Enterprise Mixed Ownership Reform Fund Co., Ltd.	0.79
Industrial and Commercial Bank of China Limited– Huitianfu CSI New Energy Vehicle Industry Index Founders Securities Investment Fund (LOF)	0.79
Citibank, National Association	0.77
China Construction Bank Corporation– Fullgoal CSI New Energy Vehicle Index Securities Investment	0.62
Wang Aijun	0.55
Guangdong Technology Risk Investment Co., Ltd.	0.52

Source: GEM Co. Ltd

GEM's parent company is Shenzhen Huifengyuan Investment Co., Ltd. which holds 8.40 percent of the company's shares. GEM founders Professor Xu Kaihua and Wang Min are the controlling shareholders of Huifengyuan Investment, holding 60 percent and 40 percent of the company's shares respectively. With 5,131 shares, assets of RMB 19.5 billion, and 10,000 employees, GEM trades its shares on the Shenzhen Stock Exchange (stock code: 002340) and SIX Swiss Exchange.

[69] GEM Co., Ltd. 2024. *2024 Semi-Annual Report.*

To ensure the security of nickel supply chain, GEM is progressing efforts to develop battery-grade nickel projects in Indonesia. QMB, which produces MHP, is GEM's first industrial processing project in Indonesia. To establish a foothold in the global vehicle electrification supply chain, GEM has expanded its nickel projects in Indonesia (see Figure 13). In partnership with MBMA, GEM is developing a new HPAL project through PT ESG New Energy Material (ESG) in which MBMA and GEM hold 60 percent and 40 percent stake respectively. The project is targeted to have a manufacturing capacity of 30,000 t/y of MHP. Phase I, with a production capacity of 20,000 t/y of MHP, is scheduled for commissioning by the end of 2024, and Phase II, with an output of 10,000 t/y, is expected to begin operations in mid-2025. The project progress has reached 81.4% as of August 2024. The company has installed a high-pressure autoclave with a capacity of 1,168 m^3, which is claimed to be the largest in the world.[70] Based on an agreement between MBMA and GEM, SCM will supply 70,000 t/y of limonite to ESG for the next 20 years.[71]

By 2030, GEM aims to produce 200,000 t/y of nickel and 18,000 t/y of cobalt at IMIP. By 2026, with a total investment of USD 3 billion, GEM expects its hydrometallurgical facilities to reach an annual potential output of 150,000 tons of nickel and 12,000 tons of cobalt.[72] To achieve this target, GEM is preparing two new projects through PT Meiming Energy Materials and Green AGCO New Energy Materials Co., Ltd which will each produce 22,000 t/y of MHP. The Environment Impact Assessment of these projects have already been approved. Additionally, through PT Qingmei Energy Materials, GEM plans to establish an integrated battery materials production chain at IMIP, with a production volume of 30,000 t/y of ternary precursor materials, 10,000 t/y of nickel carbonate (NiCO3), 5,000 t/y of lithium hydroxide (LiOH), and 11,500 t/y of semi-finished nickel. The project is scheduled to begin operations next year.[73] Most recently, GEM and Vale Indonesia have inked an agreement to build a HPAL plant in Central Sulawesi.[74]

70 GEM Co., Ltd. 2024. *2024 Semi-Annual Report.* Shenzhen: GEM Co, Ltd; PT Merdeka Copper Gold Tbk. 2024. *Quarterly Report: June 2024*; PT Merdeka Copper Gold Tbk. 2024 "Completing Autoclave Installation, PT ESG Targets HPAL Project Commissioning by the End of 2024". Available at https://merdekacoppergold.com/en/completing-autoclave-installation-pt-esg-targets-hpal-project-commissioning-by-the-end-of-2024/. Accessed July 30, 2024;

71 GEM Co., Ltd. 2024. *2024 Semi-Annual Report.*

72 GEM Co., Ltd. 2023. "Congratulations on Maiden Shipment of MHP Products in GEM's Indonesia QMB Nickel Resources Project." Available at https://en.gem.com.cn/CompanyNews/info_itemid_7104.html. Accessed July 8, 2024

73 *Ibid*.

74 Reuters. 2024. "China, Indonesia seal $10 billion in deals focused on green energy, tech". *Reuters*, November 20. Available at https://www.reuters.com/world/asia-pacific/china-indonesia-enhance-ties-with-key-deals-lithium-green-energy-tourism-2024-11-10/. Accessed November 13, 2024.

Figure 13. Business Arm of GEM Co. Ltd in in Indonesia

QMB started construction of the HPAL project on January 11, 2019. Luhut Binsar Pandjaitan, Coordinating Minister for Maritime Affairs and Investment, attended the groundbreaking ceremony. Phase I of the project with a production capacity of 31,800 t/y of MHP started operation on 26 September 2022. The Phase I will reach its peak production of about 40,000 tons of MHP by 2024. The Phase II project was commissioned in October 2022 with a production capacity of 43,000 t/y of MHP. In September 2024, the Phase II project began operating, increasing output to 65,000 t/y.[75] With a total investment of US$1.6 billion, the QMB project is targeted to have a supply capacity of more than 100,000 t/y of MHP. Employing around 2,900 workers, the company is in a good financial position (see **Figure 14**).

To maintain supply chain security, QMB cooperates with various companies. In the upstream, to ensure the supply of nickel ore, QMB has entered into a cooperation agreement with HM to supply 1.5 million tons of nickel with metal content for 20 years.[76]

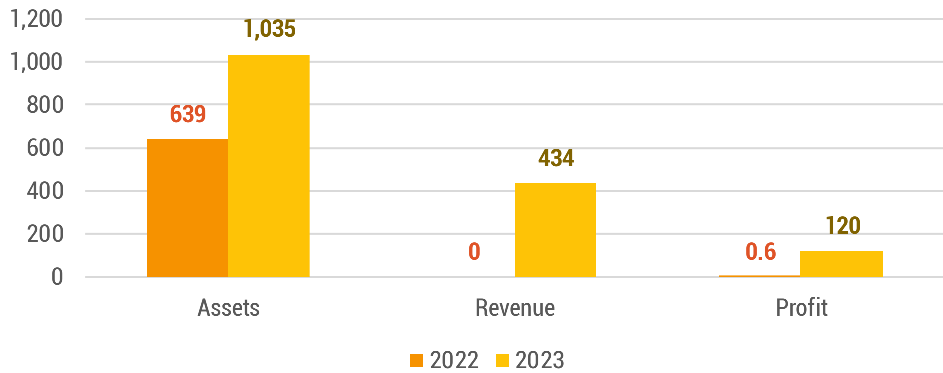

Figure 14. QMB New Energy Materials: Financial Performance in 2022 and 2023 (in Million USD)

Source: Processed from GEM[77]

75 GEM Co. Ltd. 2024. *The Third Quarterly Report of 2024.* Available at https://en.gem.com.cn/uploadfiles/2024/10/20241029220940795.pdf. Accessed September 20, 2024.

76 PT QMB New Energy Materials. 2023. *2022 Environmental, Social, and Governance (ESG) Report*. Morowali: PT QMB New Energy Materials.

77 GEM Co., Ltd. 2024. 2023 Annual Report.

FOURTH SECTION
COSTS OF UPSTREAM SUPPLY CHAIN

1. Environmental Concerns

Nickel is one of critical materials in energy transition and is substantial for EV batteries. But nickel production comes with a price. Rampant deforestation, large volume of toxic tailings, and widespread pollution caused by coal-fired power are paradoxes of the transition.

1.1. Deforestation

SCM and HM have concession areas that are classified almost entirely as forests. Of SCM's concession area, almost 98 percent, or around 20,670 hectares, is forest, including Fixed Production Forest (80.38%), Limited Production Forest (18.05%) and Protected Forest (0.01%) In the case of HM's concession area, 85.58 percent is forest, specifically Limited Production Forest (see **Figure 15** and **Figure 16**).

Deforestation is inevitable in mining areas with forest status. In the period between 2020 and 2023, Mighty Earth states that forest cover loss in SCM's area reached 661.94 hectares.[78] Since 2013, it is estimated that around 1,800 hectares of natural forest have been and will potentially be lost in HM's area, where nickel mining has been going on for more than 10 years. This estimate is based on the issuances of the first ***Borrow-to-Use Forestry Permit*** covering 851 hectares in 2013 and the second ***Borrow-to-Use Forestry Permit*** covering 994 hectares in 2019.[79]

Deforestation will accelerate in the region following the construction of IKIP inside SCM's concession area in Routa. The development of the industrial park, owned by Merdeka Group and Tsingshan Group, requires the clearing of more than 3,854.37 hectares of forest. In this regard, the Government of Indonesia already approved to clear the forest with a legal status called Production Forest Area for the Indonesia Konawe Industrial Park in February 2023.[80]

[78] Mighty Earth, *From Forest to Electric Vehicles.*

[79] Nickel Mines. 2019. *Annual Report*.

[80] PT Merdeka battery Materials Tbk. 2024. *Annual Report 2023*.

The planned expansion of IMIP's investment, which is also controlled by Tsingshan Group, will trigger rapid and widespread deforestation as it requires the clearance of thousands of hectares of forest. In March 2023, an Integrated Team formed by the Ministry of Environment and Forestry issued a recommendation for the industrial park to get clearance approval for cutting 2,156 hectares of forest area. The area includes 784 hectares of Limited Production Forest and 1,372 hectares of Convertible Production Forest.[81]

Deforestation threatens biodiversity. Mining operations are thought to have disrupted anoa (*Bubalus depressicornis*) habitat. *The International Union for Conservation of Nature (IUCN)* classifies this animal, endemic to Sulawesi, on the *red list of* endangered animals.[82] The head of the Southeast Sulawesi Natural Resources Conservation Agency stated that the forest area in the SCM mine site is the anoa's home range. Based on animal tracks, the Agency said about 15 anoa were in the area. In early July 2024, an amateur video captured an anoa being confused around the SCM employee base camp in Routa. Previously, in early June 2022, two anoa (a 7-year-old mother and a 6-month-old cub) were also caught on video in the SCM operations area. The head of Southeast Sulawesi Natural Resources Conservation Agency insisted the anoa appeared because their habitat has been disturbed by mining activities. Apart from anoa, SCM's concession area is also home to Sulawesi's endemic monkeys (*macaca*), although the population size remains unidentified.[83]

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81 Anonymous. 2023. "Pemkab Morowali Hadiri Vicon Hasil Penelitian Tim Terpadu Dalam Rangka Pengembangan Kawasan Industri PT IMIP". March 8. Available at https://morowalikab.go.id/home/read/pemkab-morowali-hadiri-vicon-hasil-penelitian-tim-terpadu-dalam-rangka-pengembangan-kawasan-industri-pt-imip. Accessed July 30, 2024.

82 See IUCN. nd. "Lowland Anoa". Available at https://www.iucnredlist.org/fr/species/3126/46364222. Accessed August 2, 2024.

83 Anonymous. 2024. "BKSDA dan Perusahaan Tambang Gelar Pertemuan Pasca-video Viral Anoa, Ini yang Disepakati". Tempo.co. July 24. Available at https://tekno.tempo.co/read/1892606/bksda-dan-perusahaan-tambang-gelar-pertemuan-pasca-video-viral-anoa-ini-yang-disepakati. Accessed August 2, 2024; Nadhir Attamimi. 2024. "Anoa Muncul di Kawasan Tambang di Konawe, BKSDA Sultra Akan Evakuasi". *Detik.com*, July 12. Available at https://www.detik.com/sulsel/berita/d-7435575/anoa-muncul-di-kawasan-tambang-di-konawe-bksda-sultra-akan-evakuasi. Accessed August 2, 2024; Annisa Aprilia Monoarfa. 2022. "BKSDA Sultra Lakukan Identifikasi 2 Ekor Anoa di Lokasi Tambang". *Tirtamedia*, June 6. Available at https://tirtamedia.id/read/bksda-sultra-lakukan-identifikasi-2-ekor-anoa-di-lokasi-tambang. Accessed August 2, 2024; Anonymous. 2022. "Habitat Terganggu, Anoa Berkeliaran Di Lokasi Tambang Nikel Di Konawe". *Kendari24.com*, June 7. Available at https://www.kendari24.com/habitat-terganggu-anoa-berkeliaran-di-lokasi-tambang-nikel-di-konawe/. Accessed August 2, 2024.

Forest Area in WUIP of PT Hengjaya Mineralindo

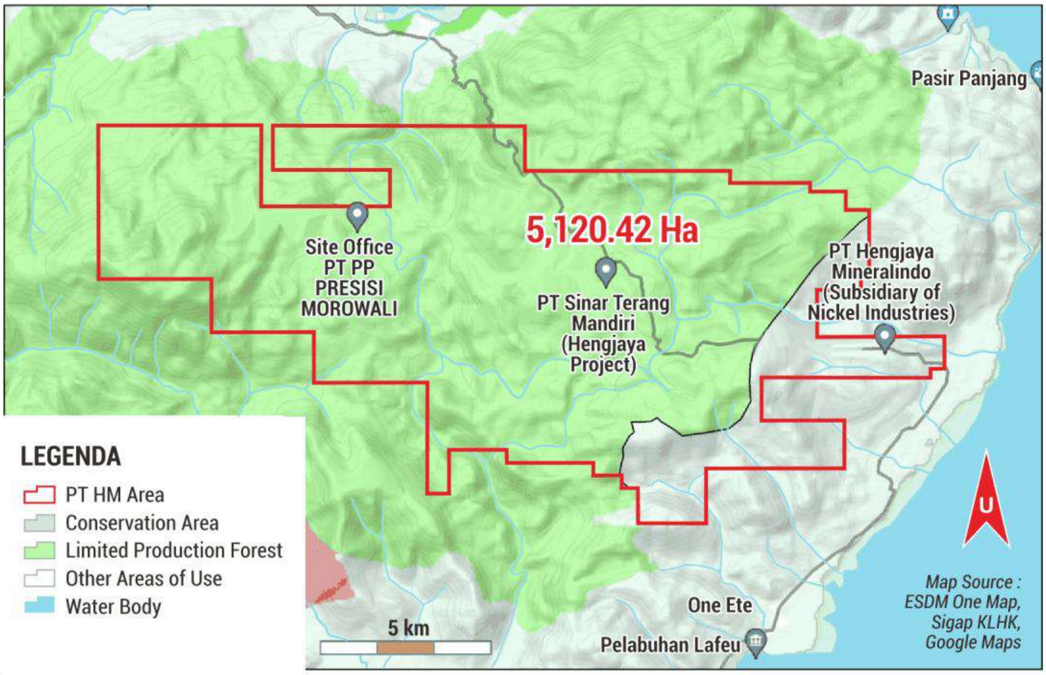

Figure 15. PT Hengjaya Mineralindo: Forest Area (hectares) within Mining Business License Area for Performing Production.

Forest Area in WUIP of PT Sulawesi Cahaya Mineral

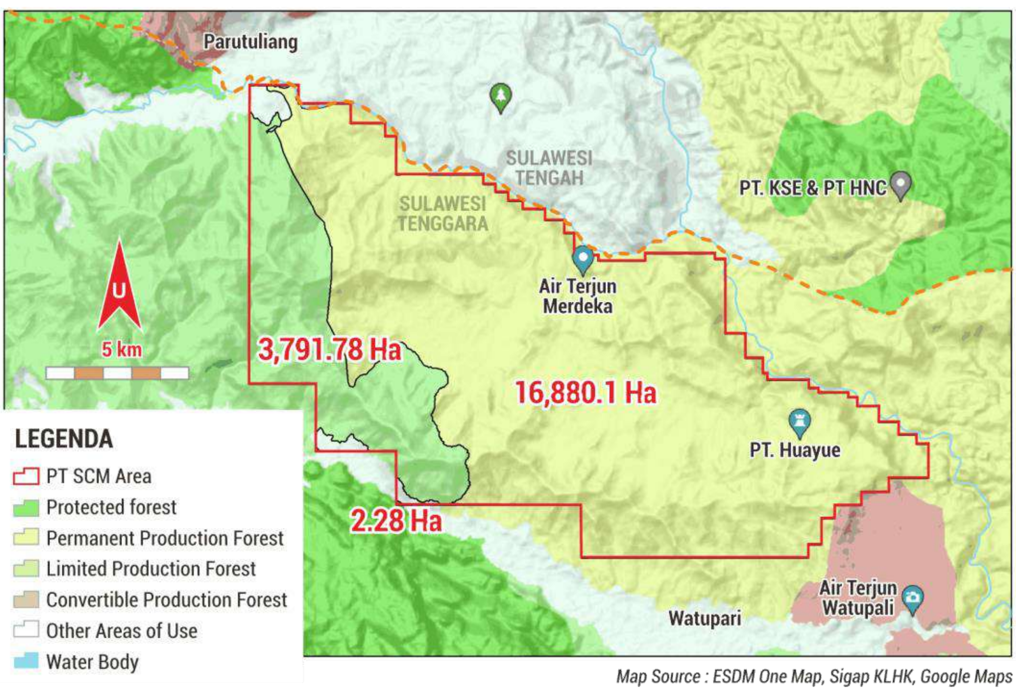

Figure 16. PT Sulawesi Cahaya Mineral: Forest Area (hectares) within Mining Business License Area for Performing Production

Sources from the Central Sulawesi Forestry Office report that the forest within the HM concession area and its surroundings is home to several endemic flora and fauna of Sulawesi. The IUCN classifies them in the Red List including ebony (*Diospyros celebica*) categorized as vulnerable as well as the Sulawesi hornbill (*Rhyticeros cassidix*) and the Sulawesi dwarf cuscus (*Strigocuscus celebensis*) which are experiencing population declines.[84]

Figure 17. PT Hengjaya Mineralindo: Mine site in Forest Areas along Trans Sulawesi between Bete Bete Village and Tangofa Village in Morowali

Source: Bram & Anto Sangadji (2024)

[84] IUCN.nd. "Diospyros celebica." Available at https://www.iucnredlist.org/search?query=Tarsius%20tatsier&searchType=species. Accessed August 2, 2024; IUCN. nd."Rhyticeros cassidix". Available at https://www.iucnredlist.org/search?query=Tarsius%20tatsier&searchType=species. Accessed August 2, 2024; IUCN.nd. "Strigocuscus celebensis". Available at https://www.iucnredlist.org/search?query=Tarsius%20tatsier&searchType=species. Accessed August 2, 2024.

Figure 18. PT Sulawesi Cahaya Mineral: Mine site in Forest Areas Around Mopute Village in Routa District, Konawe Regency

Source: Bram & Anto Sangadji (2024)

Deforestation due to nickel mining operations leads to organic material from soil flowing into the sea via rivers. Residents who rely on traditional fishing livelihoods are adversely affected as experienced by local population in Bete-Bete village where PT HM operates. During the rainy season, the seawater turns brownish resulting in a decline in fish catch. In 2021, due to PT HM's activities, residents also faced a two-month clean water shortage. During that time, the clean water reservoir near the spring was contaminated with yellow-coloured water mixed with mud. Using jerry cans and water tank trucks, residents were forced to collect clean water from a more distant source. Some residents had to buy water in gallons from refill stations.[85]

[85] Interview with Andi, a Bete-Bete resident, August 3, 2024; See also Saiful Rizal Yunus. 2024. ""Kubangan" Limbah di Tepi Bahodopi". *Kompas*, February 21. Available at https://www.kompas.id/baca/nusantara/2024/02/21/kubangan-limbah-di-tepi-bahodopi?open_from=Search_Result_Page. Accessed August 2, 2024.

Deforestation caused by open-pit nickel mining operations as a trigger for flooding has been widely reported since the beginning of the last decade.[86] In early May 2024, flash floods paralyzed the coastal areas of North Konawe Regency due to the overflow of the Lalindu River. The Trans-Sulawesi connecting the regency to Morowali Regency was completely cut off, forcing around a thousand residents to evacuate. Dozens of nickel mining companies in North Konawe Regency were considered to be the cause of the flooding.[87] In addition, in the upstream area of the Lalindu River in Routa District, 10 nickel mining companies including PT SCM have been operating with a total concession area of 47,110.2 hectares. Indeed, the forest loss caused by SCM operations and the construction of nickel-based industrial park in Routa has contributed to flooding associated with the Lalindu River. SCM itself admitted to discharging over 800,000 tons of mining and domestic wastewater into Lalindu River in 2022.[88]

1.2. Tailings

Nickel extraction using HPAL technology produces a large volume of toxic waste. In their official documents, both HNC and QMB state that they each generate 6.4 million tons of tailings per year.[89] If producing 1 ton of nickel metal generates 100 tons of tailings, then by producing 70,000 tons of MHP (Mixed Hydroxide Precipitate) in 2023, HNC has produced 7 million tons of tailings. In the same year, if QMB produced 55,000 tons of MHP according to its production capacity, approximately 5.5 million tons of tailings were generated by the company. In 2023, around 12.5 million tons of waste containing iron and manganese were sent to storage facilities, which both companies refer to in their Environmental Impact Assessment documents as "tailings dams".[90]

Nickel Mining Waste Polluting River. Image Source: AEER

[86] Sangadji, et al, *Road to Ruin*.

[87] Saiful Rijal Yunus. 2024. “Diterjang Banjir, Trans-Sulawesi di Konawe Utara Lumpuh”. *Kompas*, 3 May. Available at https://www.kompas.id/baca/nusantara/2024/05/03/diterjang-banjir-trans-sulawesi-di-konawe-utara-lumpuh. Accessed August 2, 2024; Saiful Rijal Yunus. 2024. “Banjir Berulang dan Warga yang "Terjebak" Kritisnya Lingkungan Konawe Utara”. *Kompas*, May 11. Available at https://www.kompas.id/baca/nusantara/2024/05/11/banjir-berulang-dan-dampak-kronis-lingkungan-di-konawe-utara. Accessed August 2, 2024.

[88] Merdeka Copper Gold Tbk. 2023. 2022 *Sustainability Report.* Jakarta: Merdeka Copper Gold Tbk.

[89] PT QMB New Energy Materials. 2019. *Amdal: Rencana Pembangunan Pabrik Nikel SUlfat Bahan Kimia Tingkat Baterai Kapasitas 50.000 ton per Tahun Kec. Bahudopi Kabu. Morowali Prov. Sulawesi Tengah*

[90] *Ibid*

Tailings have been one of the major issues since the planning phase of the HPAL project in IMIP. Initially, the companies planned to dispose of the waste in the deep sea (deep sea tailing placement) in Tolo Bay. However, after the government prohibited this method, the companies opted for on-land storage. HNC and QMB adopted the "dry-stack tailings" method, which involves neutralization, filtration, and compaction processes to reduce water content. The so-called dry tailings produced by this method have a moisture level of 31 percent in the case of HNC,[91] The tailings are then transported and deposited in storage facilities or tailings mounds. As the industrial park operator, IMIP provides these facilities.[92] Hamid Mina, Managing Director of IMIP, stated that nearly 600 hectares, or approximately 20 percent of IMIP's total area, have been allocated for tailings storage, with a plan to expand this area if the industrial park is developed to 6,000 hectares.[93] The distance between the HPAL plants and the storage sites is around 8–10 kilometers.[94]

Companies recognize that on-land tailings storage is considered a risk-method. Tailings storage facilities such as dams or tailings containment facilities potentially generate disaster especially in the region that is prone to earthquake, heavy rainfall, and landslide. The Spatial Planning for Morowali Regency (2019–2039) identifies Bahodopi District, where IMIP operates, as a "disaster-prone area" for earthquakes, landslides, and floods.[95] The industrial park is located on the Matano Fault in the Geresa Segment, which is an active fault and part of the Central Sulawesi Fault System (CSFS), alongside the Palu-Koro Fault.[96] Strong earthquakes have repeatedly struck Morowali in the Geresa Segment in recent years (see **Table 6**). An Earthquake, on January 4, 2021, with a magnitude of 4.9 resulted in one injury, three severely damaged houses, 26 heavily damaged worker dormitories, some damage to IMIP office facilities, and the collapse of the ceiling at an elementary school in Keurea Village. Another damaging earthquake occurred on May 31, 2024, injuring five people and causing damage to several buildings, including houses, worker dormitories, and places of worship. All victims and damages were recorded in the villages of Fatufia, Bahodopi, and Bahumakmur, near IMIP.[97]

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91 Huayou Cobalt. 2024. 2023 Environmental, Social, and Governence (ESG) Report. Huayou Cobalt.

92 Decree of the Governor (*SK Gubernur*) of Central Sulawesi No. 660/045/KLH/DPMPTSP/2020; PT Huayue Nickel Cobalt. 2019. Analisa Dampak Lingkungan; PT QMB. Analisa Dampak Lingkungan.

93 Annie Lee. 2023. "Morowali's nickel is a mainstay but highlighted as environmentally unfriendly". Bloomberg News, July 25. Available at https://www.bloombergtechnoz.com/detail-news/10945/nikel-morowali-jadi-andalan-ri-tapi-disorot-tak-ramah-lingkungan/2. Accessed August 22, 2024.

94 Interview with B9, a HNC worker and B10, a QMB workers August 15, 2024.

95 Pemerintah Kabupaten Morowali. 2019. *Rencana Tata Ruang Wilayah (RTRW) Kabupaten Morowali (2019-2039)*. Available at https://www.morowalikab.go.id/portal_morowali/file/668e15581d0f9-RKPD%20KAB.%20MOROWALI%20TAHUN%202023.pdf. Accessed September 2, 2024.

96 Adi Patria, Anggraini Rizkita Puji, Hiroyuki Tsutsumi. 2023. "Neotectonics of the eastern Matano fault, Sulawesi, Indonesia: Preliminary results". IOP Conf. Ser.: *Earth Environ. Sci*. 1227 012001; Olivier Bellier et al.2021. "High slip rate for a low seismicity along the Palu-Koro active fault in central Sulawesi (Indonesia)." *Terra Nova*, 13, 463-470.

97 Tribowo Kiswinarso dkk. 2024. Katalog Gempabumi Signifikan & Merusak Tahun 1821-2023. *Jakarta: BMKG Earthquake and Tsunami Center*; Anonymous. 2024. "Kapolres Morowali: Agar Tetap Tenang dan Tidak Terpengaruh Oleh Isu". *Tribratanews*, June 2. Available at https://tribratanews.sulteng.polri.go.id/2024/06/02/kapolres-morowali-agar-tetap-tenang-dan-tidak-terpengaruh-oleh-isu/. Accessed August 2, 2024.

Table 6. Several Earthquake Records in Morowali 2012–2024

Day/Month/Year (WITA Time / GMT +8)	Magnitude	Epicenter	Hypocenter (km)
31/05/2024 (01:08:19)	4.8	2,74°LS; 122,15°BT	20.0
19/07/2023 (18:47:53)	5.1	2,41°LS; 122,69°BT	10.0
05/03/2023 (17:05:25)	4.8	2,67°LS; 121,98°BT	5.0
04/01/2021 (03:13:59)	4.9	-2,8°LS; 122,2°BT	10.0
17/06/2020 (19:10:22)	4.8	2,75°LS; 122,18°BT	22.0
12/04/2019 (20:12:15)	5.2	1,90°LS; 122,56°BT	10.0
12/04/2019 (19:40:51)	6.8	1,89°LS; 122,57°BT	10.0
24/05/2017 (17:10:15)	5.6	2,78° LS; 122,20°BT	12.0
17/06/2017 (22:23:51)	5.1	2,63°LS; 121,73°BT	10.0
06/04/2012 (10:17:51)	5.8	-2,62° LS; 121,88°BT	26.6

Source: Processed from Meteorological, Climatological, and Geophysical Agency (BMKG) data

The Environmental Impact Assessment documents of HNC and QMB mention "dams" in relation to tailings storage facilities. However, neither company reports these in their annual Environment, Social, and Governance (ESG) documents. This differs from the HPAL project on Obi Island, which claims that its "dry stack" facility complies with international standards for large dams, such as those set by the Australian National Committee on Large Dams (ANCOLD), the International Committee on Large Dams (ICOLD), and the Global Industry Standard on Tailings Management (GISTM). The storage facility is also claimed to align with the Indonesian Dam Safety Commission.[98] It is important to state that earthquake-resistant dam construction standards are crucial in building a tailings processing facility.

Morowali has a very wet climate with an abundant rainfall. In recent years, the highest recorded rainfall in Morowali was 333 mm³ during 11 rainy days in November 2022 and 332 mm³ during 16 rainy days in March 2023. May 2023 was the longest period of consecutive rainy days.[99] The topography and land conditions around IMIP consist of red soils that are highly susceptible to hydrological activity, leading to flooding during periods of heavy rainfall. Rainfall, in addition to earthquakes, has the potential to cause dam failures if not properly accounted for. Indonesia needs to learn from human-made disasters associated with tailings mismanagement in mining industry such as the failures of the Samarco and Brumadinho tailings dams run by Vale in Brazil, which collectively claimed up to 300 lives.[100]

98 PT Halmahera Persada Lygend. 2023. *Sustainability Update Report 2023*. Jakarta: HPL

99 Pemerintah Kabupaten Morowali. 2023. *RKPD Kabupaten Morowali 2023*. Morowali:

100 Dom Phillips, "Samarco dam collapse: one year on from Brazil's worst environmental disaster"; Katy Watson, "'Vale ended our lives': Broken Brumadinho a year after dam collapse."

1.3. Coal Fired Power Plant

The energy source powering the nickel smelting and refining in IMIP comes from captive coal-fired power plants. These plants supply electricity to 54 factories within the industrial park that produce NPI, nickel matte, MHP, stainless steel, carbon steel, ferrochrome, electrolytic aluminum, graphite, lithium hydroxide, lithium carbonate, and more. This includes the electricity supply for the HPAL plants run by HNC and QMB.[101]

Global Energy Monitor recorded 3,980 MW of coal-fired power plants already operating in IMIP. However, this figure appears to be significantly lower than the actual capacity. In early 2024, Hamid Mina stated that coal-fired power plants with an installed capacity of 5,319 megawatts were already operational there. To meet the growing demand, he noted that additional coal-fired powers with an installed capacity of 1,520 megawatts were under construction. Over the past five years, the installed capacity of coal-fired power generations in IMIP has surged from around 2,000 MW in 2019 to nearly 7,000 MW today.[102] IMIP, therefore, stands as one of the most significant sources of industrial pollution in the country.

Workers and residents face pollution threats from nickel smelting and refining powered by coal-fired power plants in IMIP. QMB individually claimed that its emissions of sulfur dioxide, nitrogen oxides, and particulate matter in 2023 were approximately 248 tons, 28 tons, and 58 tons, respectively.[103] Previous studies have documented the impact of air pollution from coal-fires power and nickel smelting operations on workers and nearby residents when the installed capacity was still around 2,000 MW. At that time, official data from the IMIP clinic recorded 26,133 patient visits for chronic rhinitis.[104] Residents have repeatedly raised concerns about the effects of air pollution. Most recently, in mid-July 2024, residents of Labota Village staged a protest by blocking access to the dedicated jetty terminal and captive coal-fired power plants. They were the most severely affected by the adverse impacts of several coal-fired power and smelter operations including those run by QMB, located in their village.

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101 QMB New Energy Materials. 2024. *2023 Environmental, Social and Governance (ESG) Report*. Morowali: QMB New Energy Materials; PT Huayue Nickel Cobalt. 2022. *2021 Environmental, Social, and Governance (ESG) Report.* Morowali: PT Huayue Nickel Cobalt.

102 Global Energy Monitor. nd."Sulawesi Mining Power Station". Available at https://www.gem.wiki/Sulawesi_Mining_power_station. Accessed July 20, 2024; Global Energy Monitor. nd."Sulawesi Labota Power Station". Available at https://www.gem.wiki/Sulawesi_Labota_power_station. Accessed July 20, 2024; Anonymous. 2024." "Melihat Langsung Hilirisasi Produk Nikel di Kawasan PT IMIP, Morowali" Available at https://kumparan.com/kumparanbisnis/melihat-langsung-hilirisasi-produk-nikel-di-kawasan-pt-imip-morowali-2215p7lIUoK/full. Accessed July 20, 2024; Arfyana Citra Rahayu. 2023. "Tak Lagi tambah Tenant, Ini Penjelasan CEO Indonesia Morowali Industrial Park (IMIP)". *Kontan*, 4 October 2023. Available at https://industri.kontan.co.id/news/tak-lagi-tambah-tenant-ini-penjelasan-ceo-indonesia-morowali-industrial-park-imip. Accessed July 20, 2024

103 QMB New Energy Materials, *2023 Environmental, Social and Governance (ESG) Report.*

104 Sangadji, et al, *Road to Ruin*.

There is no indication that IMIP will reduce its use of coal-fired power plants and quickly shift to clean energy. Although IMIP plans to build a 60 MW of hydropower utilizing the Lalindu River in Konawe, there is no certainty regarding its construction timeline. More recently, IMIP announced a plan for a phased construction of a photovoltaic solar power plant starting in 2024, with an installed capacity of 1,000 MW by 2029. The solar power will require 1,000 hectares of land in mountainous areas, approximately 8 kilometers from the industrial park. As with Tsingshan's unfulfilled promise in 2021, the realization of this project is also uncertain.

2. Land Disputes

Land disputes are often inevitable in resource-based industries like nickel mining. These disputes primarily occur between villagers who depend on traditional farming activities for their livelihoods and nickel mining companies.

2.1. Bete-Bete Case

HM often associates with issue of disputes with local people arising over ownership, access, and usage of land such as in Bete-Bete Village in Bahodopi District. HM legally controls the nickel resources in the village called the Bete-Bete Block, estimated to contain 5.5 million tons with 1.9% nickel and 0.04% cobalt content, and the Bete-Bete West Block, which has 1.2 million tons with 1.8% nickel and 0.05% cobalt content. HM firstly mined in these blocks, which are only about 12 kilometers from IMIP, where villagers assert ownership of lands based on long-standing customs and practices.[105]

The dispute erupted when the company encroached on the plantations of 30 residents, primarily from 2017 to 2020. HM built hauling roads and mined on productive pepper and clove plantations. Unable to endure the situation, a resident of Bete-Bete burned down HM's field office in 2018 in this village. He was reported to the police on charges of destroying company property.[106]

105 Nickel Mines Limited. 2019. *2018 Annual Report*. Sydney: Nickel Mines Limited; Patersons Securities Limited. 2019. Research Notes.

106 Interview with Andi, a Bete-Bete resident, August 3, 2024

Villagers had frequently reported land grabs to the Regional People's Representative Council of and the local government agencies, but the issue remained unresolved for years. On July 3, 2020, they blockaded the company's hauling road, demanding compensation for land and crop damage. The company later removed the blockade covertly without negotiating with the residents. It was reported that the company did so by requesting security assistance from the Morowali Police Department.[107]

Figure 19. Prohibition of Entry into the Hengjaya Mineralindo Concession Area

[107] Idham. 2020. "Diam-diam, PT Hengjaya Mineralindo Buka Palang Warga Desa Bete-Bete". 24 September. Available at https://kailipost.com/2020/09/diam-diam-pt-hengjaya-mineralindo-buka-palang-warga-desa-bete-bete.html. Accessed July 5, 2024.

Unwilling to accept the situation, residents protested on October 13, 2020. The protest resulted in an agreement to hold a meeting between the residents and HM a week after the protest. The company requested that the Government of Morowali facilitate the meeting.[108] The government then sent a letter to the village head to arrange the meeting with an agenda focused on corporate social responsibility. However, the residents refused to attend, arguing that the meeting agenda was unrelated to their demands. They then resumed their blockade of the hauling road, halting mining operations between October 20 and October 28, 2020. Following the protest, five Bete-Bete villagers, including the village head, were arrested by the police on October 29, 2020. On January 21, 2021, the Central Sulawesi Regional Police issued detention orders for Ridwan (Village Head), Sahrir (Chair of the Village Consultative Body), Sofyan, Lukman, and Rivai. They were accused of threatening and extorting a Hengjaya employee, Jerry Charly, in addition to the road blockade.[109]

2.2. Lafeu & Tandaoleo Case

Land disputes within HM's concession area have also occurred with villagers at Lafeu and Tandaoleo in Bungku Pesisir District. The residents claim that they had been engaging in farming and plantation activities long before HM's arrival. A total of 73 residents from Lafeu and 70 residents from Tandaoleo assert ownership of these lands. On the other hand, HM claims that the lands are part of its concession area.

Disputes were inevitable. On December 22 and 23, 2023, residents of both villages staged a protest. They demanded compensation for damaged crops and blockaded the hauling road to HM's jetty, halting nickel ore transportations.

> "We are citizens and part of this nation. We work the land to survive, not for profit. Our husbands are old and unlikely to be hired by the company. We only depend on the yields from our palm oil trees, other crops, and vegetables. But our crops were destroyed without any compensation. We are being oppressed by Hengjaya. Our land is being taken away. But don't give up," said Sunarti during her speech.[110]

108 Patar Juphian. 2020. ""PT.HM Abaikan Hak-Hak Warga Desa Bete Bete". *Nuansa Pos*, 3 November. Available at https://nuansapos.com/pt-hm-abaikan-hak-hak-warga-desa-bete-bete/. Accessed July 5, 2024.

109 Interview with Andi. See also Patar Jupian. 2021. "Bela Hak Warganya Kades Bete-bete Dkk di Penjarakan PT.HM, PEMKAB Morowali Terkesan Biarkan". *Nuansa Pos*, 31 January. Available at https://nuansapos.com/bela-hak-warganya-kades-bete-bete-dkk-di-penjarakan-pt-hm-pemkab-morowali-terkesan-biarkan/. Accessed 3 July 2024; Anonymous. 2020. "Warga Digiring Paksa, Puluhan Warga Bete-Bete Datangi Polres Morowali". *Posonews*, October 30. Available at https://posonews.id/2020/10/30/warga-digiring-paksa-puluhan-warga-bete-bete-datangi-polres-morowali/. Accessed July 3, 2024; Patar Jupian. 2021. ""PT.HM Penjarakan Kades Bete-bete dan Sejumlah Warganya, Sofyan : Ini Kriminalisasi". *Nuansa Pos*, 20 January. Available at https://nuansapos.com/pt-hm-penjarakan-kades-bete-bete-dan-sejumlah-warganya-sofyan-ini-kriminalisasi/. Acessed 3 July 2024.

110 Anonymous. 2023."Wakil Masyarakat Dua Desa – Duduki Jalan Hauling PT. Hengjaya Mineralindo". *Cakrabhayangkaranews*, 26 December. Available at https://cakrabhayangkaranews.com/wakil-masyarakat-dua-desa-duduki-jalan-hauling-pt-hengjaya-mineralindo/. Accessed July 3, 2024.

The protest drew the attention of the Government of Morowali. On February 18, 2024, Acting Regent of Morowali, Rahmansyah Ismail acknowledged that the Morowali government had formed a team to verify the lands and crops of residents that had not yet been compensated. He insisted that the company must be responsible for unresolved land issues and expressed no objection if residents protested the company. He even argued that the company's operations should be halted if HM refused to fulfill its obligations to the residents.[111]

However, no concrete resolution from the Regent materialized. On March 18, 2024, residents of both villages staged a protest in front of the official residence of Morowali's Regent. They set up tents and demanded that the Government of Morowali expedite compensation. The authorities attempted to persuade the residents to dismantle the tents and redirect the protest to the Regional People's Representative Council of Morowali. However, the residents refused arguing that the Council had failed to resolve the issue.[112]

2.3. Routa Case

Fierce land disputes have also occurred in SCM's concession site in Routa District, Konawe Regency, Southeast Sulawesi. Villagers accuse SCM of seizing their lands and destroying plantation crops. These disputes have been marked by protests and demonstrations by villagers over the past few years. The disputes are also fueled by demands from locals for SCM to employ local people in the company.

The dispute in Mopute, an area claimed by SCM as part of its concession, is one of the ongoing conflicts. Mopute is an old settlement of an indigenous group called Tolaki people and is located along the road to Lalomerui Village in Routa District. Residents consider the approximately 2,000-hectare area to be their ancestral land and ancient village, evidenced by coconut and sago trees, rice field embankments, and ancestral graves.[113] The location has also been a site for resin collection by villagers for generations, from colonial times until the early 2010s.[114]

The Darul Islam/Indonesian Islamic Army (DII/TII) rebellion in the 1950s, which used Routa as one of its bases, forced the entire Routa population, including those in Mopute, to flee the area. They dispersed to regions around Asera (now North Konawe Regency) and Kendari, as

111 Anonymous. 2024. "Bupati Hadir Mewakili Negara "Terkait Konflik Antara Masyarakat Dengan PT. Hengjaya." *FileSulawesi*, 18 Februari. Available at https://filesulawesi.com/2024/02/18/bupati-hadir-mewakili-negara-terkait-konflik-antara-masyarakat-dengan-pt-hengjaya/. Accessed July 3, 2024.

112 Imam Muslik. 2024. "Tanpa Ganti Rugi, PT. Hengjaya Rusak Tanam Tumbuh Masyarakat". *Kabarluwuk*, 18 Maret. Available at https://kabarluwuk.com/tanpa-ganti-rugi-pt-hengjaya-rusak-tanam-tumbuh-masyarakat/. Acessed July 3, 2024.

113 Interview with Baharuddin Maranai in Mopute, 23 August 2024; See also Franco B. Dengo & Ian Morse. 2024." Kongsi Hilirisasi Nikel di Kawasan Industri Nikel Konawe Menggusur Hutan Adat Mopute." *Project Multatuli*, 7 February. Available at https://projectmultatuli.org/kongsi-hilirisasi-nikel-di-kawasan-industri-konawe-menggusur-hutan-adat-mopute/. Accessed August 27, 2024.

114 Iskandar. 2013. "Tenure Claims over Damar Forest Resources from the Perspective of Indigenous (Local) Communities in Mining Areas (Case of Lalomerui Village Community, Routa Sub-district)". *Bulletin of Socio-Economic Agricultural Research.* Edition No. 28, Year 15: 64-70

well as Bungku and Bahodopi (now Morowali Regency). Some residents gradually returned to their original villages after order was restored, while others remained in new settlements but continued to maintain their traditional claims over natural resources in Routa.[115] Specifically, residents of Routa in Mopute, while displaced, named their new village after Mopute, which is now part of Oheo District in North Konawe Regency.[116]

SCM's activities around the old Mopute settlement have triggered disputes. Within a radius of approximately 2.5–3 kilometers, SCM is conducting mining operations. Residents state that SCM plans to build solar panels and a waste disposal site in Mopute. The locals reject and occupy the area and built settlements there, constructing about 200 houses with wooden walls and zinc roofs as well as a place of worship and opening agricultural land. Police arrested three residents at the end of 2022 on charges of entering SCM's area. They were released after agreeing to dismantle their houses and leave the settlement. Most of the houses still stand today. Rumors have circulated that SCM paid IDR 12 billion (USD 800,000) in compensation, which was handed over to several community leaders. However, this compensation scandal has instead fueled dissatisfaction among residents, leading them to continue occupying the old Mopute settlement area.[117]

Nickel Mine in Konawe.
Image Source: Haryo Wirawan/ BBC Indonesia

115 Interview with Baharuddin Maranai; See also Andrew McWilliam. 2011. "Marginal Governance in the Time of Pemekaran: Case Studies from Sulawesi and West Papua". *Asian Journal of Social Science* 39(2): 150-170, p 159-60; About DI/TII see Barbara Sillars Harvey. 1989. *Pemberontakan Kahar Muzakkar: Dari Tradisi ke DI/TII*. Jakarta: Grafitti Press.

116 Interview with Baharuddin Maranai.

117 Interview with Baharuddin Maranai; See also Franco B. Dengo & Ian Morse," Nickel Downstream Concession in Konawe Industrial Estate Displaces Mopute Customary Forest".

Protests against SCM have even escalated into violence in Lalomerui Village. On February 15, 2023, hundreds of residents staged a demonstration to disrupt mining operations. The protest turned chaotic when the company refused to meet with the protesters and police prevented demonstrators from entering SCM's site. The crowd threw stones at police officers who were equipped with shields. One protester was injured, and the police claimed that the injury was caused by a thrown stone. However, protesters alleged that the injury resulted from a jab from a police baton. One of the triggers for the protest was the issue of employing local labor. The company claimed that 57 percent of its workers were locals, while protesters stated that only 490 local peoples were employed by the company, which reportedly has a workforce of 4,000 employees.[118]

Hundreds of residents from Routa resumed their protest by blocking the SCM hauling road in Lalomerui Village. They set up tents in the road obstructing the movement of company vehicles. After four nights of demonstrations, the protesters dispersed under several conditions. Among them, SCM promised to form a team to resolve the issue of compensation for crops. SCM and its vendors also pledged to provide opportunities for the locals to work.[119]

Routa residents block PT SCM's Hauling Road in Lalomerui Village, Routa District, Konawe. (17/1/2023). Image Source: kendariinfo.com

[118] Anonymous. 2023. "Unjuk Rasa Warga Routa Konawe di PT SCM Berakhir Ricuh, 1 Orang Luka". *Kendari Info*, 15 February. Available at https://kendariinfo.com/unjuk-rasa-warga-routa-konawe-di-pt-scm-berakhir-ricuh-1-orang-luka/. Accessed August 27, 2024; Febriyono Tamenk. 2023. "Ratusan Warga Konawe Bentrok di Perusahaan Tambang, 1 Terluka". *Sindonews*, 16 February. Available at https://daerah.sindonews.com/read/1023965/604/ratusan-warga-konawe-bentrok-di-perusahaan-tambang-1-terluka-1676480522. Accessed August 27, 2024.

[119] Anonymous. 2023. "Warga Routa Terpaksa Bermalam di Jalan Hauling PT SCM, Buntut Tuntutan Belum Terealisasi." *Kendari Info*, 15 February. Available at https://kendariinfo.com/warga-routa-terpaksa-bermalam-di-jalan-hauling-pt-scm-buntut-tuntutan-belum-terealisasi/. Accessed August 27, 2024; Anonymous. 2023. "4 Hari Blokade Jalan Hauling PT SCM, Warga Routa Akhirnya Membubarkan Diri". *Kendari Info*, 19 February. Available at https://kendariinfo.com/4-hari-blokade-jalan-hauling-pt-scm-warga-routa-akhirnya-membubarkan-diri/. Accessed August 27, 2024.

On May 22, 2024, without a clear resolution, around 40 Routa residents staged a protest at the Konawe Regent's office. The protesters demanded that the Regent asks SCM to expedite compensation for crops allegedly damaged by the company. After being received by the Konawe Regional Secretariat, the residents dispersed but threatened to block the hauling road if their demands were not met.[120]

The Regency Government of Konawe subsequently facilitated the compensation resolution process. In September 2023, the Regent claimed that an agreement had been reached between SCM and the locals. Both parties agreed to compensation of IDR 90 million (USD 6,000) per hectare for coffee crops. With a total area of 108 hectares, the company was estimated to pay IDR 9.6 billion (USD 640,000) to the residents. SCM then compensated several residents whose land overlapped with the mining area. On October 16, 2023, facilitated by the Acting Regent of Konawe, SCM paid IDR 4.6 billion (USD 306,666) to four landowners for a total area of 48 hectares. The compensation process took place at the Regent's office and was attended by security officials (police and military).[121]

Despite repeated protests and some residents receiving compensation, the issue remains unresolved. At the end of January 2024, hundreds of people from the Konawe Routa Family Forum (Fokber) staged protests at the Directorate General of Minerals and Coal – Ministry of Energy and Mineral Resources and the Investment Coordinating Board (BKPM) in Jakarta. They demanded compensation for residents' agricultural land in SCM's mining area, stating that the compensation issue had remained unclear for approximately 13 years. The protesters urged the Directorate General of Minerals and Coal and the Investment Coordinating Board to temporarily halt SCM's operations.[122]

3. Labor Issues

Labor issues in the nickel industry always stem from companies' efforts to maximize profits by exploiting cheap labor. This exploitation occurs through wage suppression, extended working hours, poor occupational health and safety standards, and unilateral actions by companies, such as salary deductions, layoffs, and imposing sanctions on workers.

[120] Anonymous. 2023. "Protest of Routa Konawe Residents at PT SCM Ends in Chaos, 1 Person Injured".

[121] Andika. 2023. "Rp 4,6 Miliar Ganti Rugi Tanaman Warga Routa". Kendari Pos, 17 October. Available at https://kendaripos.fajar.co.id/2023/10/17/rp-46-miliar-ganti-rugi-tanaman-warga-routa/. Accessed August 25, 2024; Annisa Nurdiassa. 2023."4 Warga di Kecamatan Routa Konawe Terima Uang Ganti Rugi Lahan Capai Rp4,6 Miliar dari PT SCM. *Tribun News*, 16 October. Available at https://sultra.tribunnews.com/2023/10/16/4-warga-di-kecamatan-routa-konawe-terima-uang-ganti-rugi-lahan-capai-rp46-miliar-dari-pt-scm. Accessed August 25, 2024; Abdul Azis Zenong. 2023. "Pemkab Konawe memfasilitasi ganti rugi tanaman kopi warga Routa. Available at https://sultra.antaranews.com/berita/450132/pemkab-konawe-memfasilitasi-ganti-rugi-tanaman-kopi-warga-routa?. Accessed August 25, 2024.

[122] Anonymous. 2024. "Demo di Kementerian, Ratusan Massa Tuntut PT. SCM Segera Bayarkan Lahan Masyarakat". *Teropong Sultra*. Available at https://www.teropongsultra.com/nasional/demo-di-kementerian-ratusan-massa-tuntut-pt-scm-segera-bayarkan-lahan-masyarakat. Accessed August 25, 2024.

3.1. Low Wages and Long Working Hours

The nickel processing industry in Morowali mostly use a wage system based on an extreme exploitation of labour. Workers earn higher nominal wages compared to other sectors in the region, but these higher wages result from extended working hours. Companies suppress base salaries for regular working hours and incentivize workers to earn higher wages by working longer overtime hours. The more overtime hours worked, the higher the total wages received. This practice has been ongoing for years in IMIP, and included companies like HNC and QMB, which exploit labor by paying low base salaries insufficient to meet workers' subsistence needs. Conversely, companies offer opportunities for extended overtime, allowing workers to earn higher wages. This method effectively encourages workers to compete for longer overtime hours to bring home higher earnings (see **Box 1**).[123]

However, these excessive overtime hours violate government regulations, which stipulate that overtime should not exceed 4 hours per day and 18 hours per week. Beyond this, excessive working hours infringe on fundamental human rights, such as the right to health, the right to fair working conditions, and the right to family life.[124]

Box 1. Extreme Exploitation Based on Overtime

A worker at QMB earns a take-home pay of IDR 9.9 million (USD 660) per month. This consists of **fixed components** amounting to **IDR 4.02 million (USD 268)**, including a base salary of IDR 3.1 million (USD 206), housing allowance of IDR 600,000 (USD 40), family allowance of IDR 200,000 (USD 13.33), and location allowance of IDR 100,000 (USD 6.6). **The non-fixed components** total IDR **5.83 million (USD 388.66)**, comprising overtime pay of IDR 5.8 million (USD 386.66) and a work bonus of IDR 30,000 (USD 2). The overtime pay far exceeds the fixed wage components. After deductions (1% for pensions, 2% for old-age security, absentee penalties, and other deductions), the worker takes home IDR 9.2 million (USD 613.33).[125]

Excessive overtime hours remain a critical issue. The QMB worker spends 111 overtime hours per month to earn IDR 5.8 million. This translates to overtime hours of 6–8 hours per day or 22–28 hours per week.[126]

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123 Interview with Afdal Amin, Chairperson of the Morowali Industrial Workers' Union (SPIM), August 29, 2024.

124 See Article 16, 23, 24 from Universal Declaration of Human Rights. See also United Nations. Nd. *Universal Declaration of Human Rights*. Available at https://www.un.org/en/about-us/universal-declaration-of-human-rights. Accessed August 10, 2024; See also Article 23 of the International Covenant on Civil and Political Rights. See also United Nations. Nd. "International Covenant on Civil and Political Rights". Available at https://www.ohchr.org/en/instruments-mechanisms/instruments/international-covenant-civil-and-political-rights#:~:text=in%20that%20Convention.-,Article%2023,a%20family%20shall%20be%20recognized. Accessed August 10, 2024.

125 Interview with B1, a QMB worker on August 10, 2024 and a pay slip.

126 *Ibid.*

3.2. Occupational Health and Safety

Occupational Health and Safety is a major issue in the nickel mining and processing industry. Workers involved in HPAL projects face risks of exposure to hazardous chemicals, particularly sulfuric acid, which is a critical component in nickel extraction. However, sulfuric acid is also highly dangerous when it encounters or is absorbed by the human body. Exposure can cause skin irritation, burns, and long-term health effects. Both HNC and QMB operate sulfuric acid plants at IMIP.

On March 19, 2024, a sulfuric acid gas leak occurred at PT Merdeka Tsingshan Indonesia (MTI), another company at IMIP with a sulfuric acid facility. The leak affected around 40 workers from four neighboring companies, causing symptoms of shortness of breath and dizziness. A subsequent leak occurred at QMB on August 3, 2024. IMIP officially reported that the incident was contained five hours after it occurred. Management assured worker safety, as workers were prohibited from entering the affected area. Meanwhile, residents living nearby were claimed to be unaffected due to the 1.5-kilometer distance between the incident site and the residential area.[127]

The nickel processing operation generates air pollution that poses long-term health risks to workers. Airborne pollutants containing nickel can damage internal organs over time. Emissions from the HPAL process include H_2SO_4, sulfur dioxide (SO_2), and carbon dioxide (CO_2). Toxic gases like sulfur dioxide can irritate the respiratory system. HNC claims that when its sulfuric acid plant operates normally, the SO_2 concentration in emissions is 641 mg/Nm^3. Amid these polluted working conditions, an HNC mechanic reported experiencing shortness of breath for two days. Unaware of being exposed to hazardous substances, he suspected that dust from passing material transport trucks was the cause. He also noted that the company did not provide workers with standard masks.[128]

Nickel processing facilities also operate machinery and equipment that produce loud noise. Prolonged exposure to such noise can lead to hearing impairments. A dump truck maintenance worker at HNC reported experiencing hearing issues due to excessive noise. He stated that workshop workers did not use hearing protection to mitigate the noise. The same worker also complained of lower back pain (LBP) from repetitive seated work over long periods.[129] LBP has been a prominent health issue at IMIP for years. In 2019 alone, at least 2,900 patients with LBP complaints sought treatment at the IMIP clinic.[130]

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127 Communication via Whatsapp, August 4, 2024; see also Marhum. 2024. "Heboh! Beredar Video Kebocoran Pipa Asam Sulfat Selama Tiga Hari, PT IMIP: Itu Hoaks". *Sangalu*, 3 August. Available at https://www.sangalu.com/daerah/8313267602/heboh-beredar-video-kebocoran-pipa-asam-sulfat-selama-tiga-hari-pt-imip-itu-hoaks. Accessed August 4, 2024.

128 Interview with B5, a HNC worker, August 11, 2024.

129 Interview with B4, a HNC worker, August 15, 2024

130 Arianto Sangadji et al *Road to Ruin*.

IMIP has a notoriously poor reputation related to cases of fatal workplace accidents. The most significant incident was a fire at PT Indonesia Tsingshan Stainless Steel (ITSS) on December 23, 2023.[131] Many workplace accidents go unreported to the public because the company has no intention of disclosing them. The company even pressures workers not to share information about workplace accidents. Sources among HNC workers reported at least two deaths of their colleagues in 2023 that were not made public. The first accident involved a worker driving a heavy-duty truck transporting cement for the company's construction project in Routa. The second involved a Chinese worker who was in a traffic accident in Tangofa Village when the brakes on his vehicle failed while in motion. HNC never disclosed these incidents publicly, and mainstream media also did not report them. The only information and photos were shared among workers through WhatsApp.[132] In its *2023 ESG report*, Huayou Cobalt mentioned one work-related death in 2023 but did not specifically attribute it to HNC or its other subsidiaries. Workers acknowledged that there are many more accident cases unknown to the public, where workers sustained severe or minor injuries.[133]

PT Indonesia Tsingshan Stainless Steel Smelter Furnace Explodes and Kills 13 Workers, on December 23, 2023.
Image Source: kilasjatim.com

[131] See Sembada Bersama. 2024. *The Morowali Explosion: An industrial accident that killed workers in the heartland of the nickel industry.* Jakarta: Sembada Bersama.

[132] Interview with B3 and B6, HNC workers, August 25, 2024.

[133] Interview with B3 and B6, HNC workers and with Afdal Amin August 25, 2024.

Transportation-related accidents are often caused by poor vehicle maintenance. One such incident involved a dump truck driver at HNC, who had an accident at a waste storage area on June 17, 2024. The truck he was driving fell into a hole and overturned, causing a sprained left hand and an injury to his thigh bone. Before the incident, the victim had reported to a supervisor that the vehicle was unfit for operation. However, the supervisor ignored the report and simply said, "Just use it!".[134] An HNC safety worker confirmed finding minor damage on vehicles that required repair or spare part replacement. However, many of these reports were not acted upon after being submitted to supervisors.[135]

QMB workers also complained about poor vehicle maintenance at their workplace. They reported a collision between heavy vehicles in the early morning while transporting tailings on a slippery hauling road after rain. Workers said that in addition to fatigue, the incident could have been prevented if supervisors had addressed their reports about the poor condition of the dump trucks, such as brake and tire issues. They also mentioned frequent problems with the trucks they operated, like air conditioning units that had not worked for months. As a result, workers were exposed to dust on hauling roads and production sites, while the company did not provide them with standard masks.[136]

In workplace accident cases, workers often face penalties. About two weeks after a truck accident in mid-June 2024, HNC issued a warning letter to the injured worker, who was also required to pay a fine of IDR 1.7 million (USD 113). Considering this unfair, the worker refused to sign the letter.[137] These incidents are like the tip of the iceberg: occurring frequently but failing to become a shared issue among workers. Workers often feel powerless and choose to sign the documents out of fear of further sanctions, which could lead to dismissal.[138] For example, this was the case with a QMB driver who received a warning letter after a heavy vehicle collision.[139]

[134] Robert Dwiantoro. 2024. "Nestapa pekerja di IMIP; alami kecelakaan kerja berujung sanksi ganda dari korporasi". *Tutura*, 8 July. Available at https://tutura.id/homepage/readmore/nestapa-pekerja-di-imip-alami-kecelakaan-kerja-berujung-sanksi-ganda-dari-korporasi-1720447711. Accessed August 20, 2024.

[135] Interview with B11, a HNC worker, August 25, 2024

[136] Interview with B7, a QMB worker, August 27, 2024

[137] Dwiantoro, "The plight of workers at IMIP; work accident leads to double sanctions from the corporation".

[138] Interview with Afdal Amin August 25, 2024.

[139] Interview with B8, a QMB worker August 16, 2024.

3.3. Wage Deductions

Wage deductions have often become an issue at IMIP. In June and July 2024, HNC was reported to have deducted wages, including performance bonuses, citing a decline in performance. Initially, the deductions were around IDR 50,000 (USD 3,33) but later escalated to millions of rupiah, including unpaid wages based on an Overtime Work Order. According to workers, the implementation of the Indonesian Employee Performance Management System, which governs point deductions, has impacted the reduction of Performance Allowances. They stated that management arbitrarily imposed deductions without justification or prior notification to workers.

These deductions led to disputes between workers and the company. The Head of the Workers' Union Branch of the National Workers Union (SPN) at HNC raised concerns about the issue, as it affected nearly all union members. Bipartite negotiations were held three times, but the problem remained unresolved.[140]

IMIP workers. Image Source: carigaji.com

[140] SPN News. 2024. "Managemen PT> Huayue Nickel Cobalt (HYNC) Diduga Memotong Upah Karyawan Tanpa Penjelasan Jelas". 11 August. Available at https://spn.or.id/manajemen-pt-huayue-nickel-cobalt-hync-diduga-memotong-upah-karyawan-tanpa-penjelasan-jelas/. Accessed August 20, 2024.

Indonesia is a key country for the global nickel supply chain with 42.30 percent of global nickel reserves, representing 50 percent of total mined nickel and 42 percent of processed nickel in 2023. Even though Indonesia's processed nickel is mostly being used for stainless steel production, rapid growth of Chinese investment has turned Indonesia into the new center of nickel production. This is done through High-Pressure Acid Leaching (HPAL) technology that results in mixed hydroxide precipitate (MHP), a raw material of electric vehicle (EV) batteries.

Two mining companies, Sulawesi Cahaya Mineral (SCM) Ltd. and Hengjaya Mineralindo (HM) Ltd., and two smelter companies, Huayue Nickel Cobalt (HNC) Ltd. and QMB New Energy Materials (QMB) Ltd are part of the mixed hydroxide precipitate (MHP) supply chain production at the Indonesia Morowali Industrial Park (IMIP). Because these companies are part of large holdings in the integrated nickel business made up of Huayou Cobalt, GEM Co. Ltd, Merdeka Group, Nickel Industries, and Tsingshan Group, nickel production for electric vehicle batteries in Indonesia is characterized as monopolistic.

The role of Indonesia nickel production in the global green energy transition is ironic. Nickel mining operations are mandating deforestation and threatening biodiversity, causing water pollution, and triggering a lot of lawsuits against local communities. Nickel production generates low-paying jobs and use massive amounts of fossil fuels at captive coal power plants. HM, SCM, HNC, and QMB represent the filthy face of Indonesia's nickel supply chain.

To prevent more damages, the Indonesian government and nickel companies must implement high environmental and human rights standards as mandatory requirements in the nickel mining and smelting activities.

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